

Oil Outlook 2024/2025

The Endgame

- After dropping by an estimated 17% in 2023, Brent oil price is expected to remain largely flat in 2024 and edge further down 10% in 2025. Our Brent forecast for this year and next has not changed since June and is expected to average \$81 for 2023 (YTD realized at \$82.62) and \$83 in 2024 ([The labors of Sisyphus](#), 14 June 2023). In this report we extend our outlook through 2025 with an average Brent price target of \$75/bbl.
- **World oil demand is on pace to grow a solid 1.9 mbd in 2023**, boosted by asynchronous conditions in China as it reopened from lockdowns (2/3 of demand growth), paired with reasonably resilient developed world economic conditions. More than offsetting this, non-OPEC supply surged 2.2 mbd, lead by the US (1.5 mbd), leaving OPEC+ alliance having to cut production to balance the markets.
- **In 2024, we expect more of the same.** Despite sustained economic headwinds, we see demand rising by a solid 1.6 mbd, underpinned by robust EM, resilient US and weak but stable Europe. Demand composition will likely flip, with two-thirds of demand gains set to come from the overall economic expansion, while continued normalization of jet fuel would contribute the rest. Non-OPEC+ producers again should drive overall growth in supply—projected at 1.7 mbd—more than matching growth in demand. To keep the oil market balanced, OPEC+ alliance would need to continue to constrain production.
- **Oil balances should loosen further in 2025.** Global oil demand growth will likely decelerate to 1 mbd as the last phase of the post-pandemic rebound dissipates and advancing energy efficiencies and an expanding electric vehicle fleet gain ground. Non-OPEC+ supply is set to surge by 2 mbd underpinned by large-scale offshore developments in Brazil, Guyana, Senegal and Norway. In total, eight floating production, storage and offloading (FPSO) vessels with a combined capacity of 1.4 mbd are committed for delivery in 2025. With demand gains set to slow, and non-OPEC supply surging, the market would likely shift into a large surplus. Invariably, to make room for this supply growth, OPEC+ members would need to *deepen current cuts by 1 mbd*, were the alliance to adhere to market management.
- **All this throws a sharp focus on OPEC+'s exit strategy.** We expect Saudi Arabia and Russia to extend their voluntary production/export cuts through 1Q24. Our view on demand in 2024 suggests that as long as the broader OPEC+ maintains the 2 mbd cut in quotas agreed in November 2022 and the extra 1.7 mbd of curbs from some members pledged in April 2023, Saudi and Russia will be able to *unwind* some of their 1.3 mbd voluntary reductions starting in April 2024. Assuming Saudi Arabia pumps additional 250 kbd and Russia increases its exports by 150 kbd, global oil inventories would likely stay flat in 2024 and build 1.2 mbd in 2025. In this scenario, Brent oil price should average \$83 in 2024 and \$75 in 2025.
- **However, a case could be made for choosing higher volumes in order to achieve better optionality in the future.** Were Saudi and Russia to bring back

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the full 1.3 mbd of self-imposed curbs into the market in April, global oil surplus would surge to 700 kbd in 2024 and 2.1 mbd in 2025. Our pricing model suggests that the annualized impact on the price will be \$6 in 2024 (to \$77) and \$10 in 2025 (to \$65), after adjusting for potential supply response from price-sensitive producers. Spot WTI below \$70 and future strip into low \$60s would likely trigger a slowdown in drilling and fracking activity from the US producers in 2H24, shaving 170 kbd off our 760 kbd projected growth in the US oil supply in 2024. In 2025, US oil and condensate production will likely contract by 150 kbd, while NGL output will continue to grow by 150 kbd, keeping total US liquids production flat compared to our baseline projection of a 740 kbd growth.

- **There is a substantial number of elections in 2024** (Taiwan in January, Russia in March, US in November, Venezuela late 2024, etc.) that could potentially heighten volatility of oil prices.

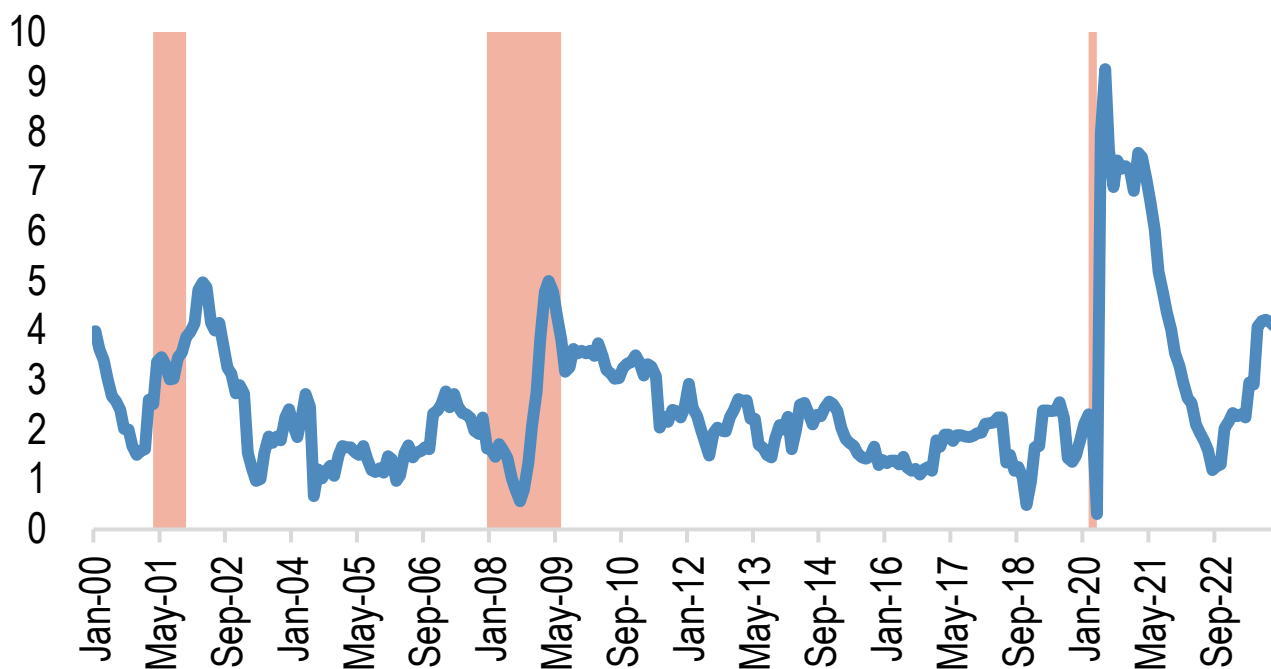
The endgame

In our 2023 Outlook last November we argued that the reality for the oil market is that there is simply more supply than demand in 2023 ([Oil Outlook 2023/2024: \\$90 redux](#), 28 November 2022). Although we had a constructive view on demand, projecting it to expand at a robust 1.3 mbd, we expected supply to grow 1.7 mbd, with non-OPEC supply alone contributing 1.6 mbd ([Ample demand, but too much supply](#), 9 May 2023). Consequently, the first iteration of our price forecast, purely based on supply and demand balances and devoid of any policy reaction, resulted in an average Brent price of \$84 in 2023. We believed the price was low enough to elicit a supply response and our price forecast of \$90 in 2023 rested on the view that OPEC+ alliance would need to do the heavy lifting to keep markets balanced. We also argued that it would be difficult for oil prices to breach \$100 this year ([\\$90 is a target, \\$100 is a stretch](#), 19 January 2023).

Defying the overall bearishness, the world's oil demand would likely grow a solid 1.9 mbd this year—almost 500 kbd above our initial estimates. Crucially, more than offsetting this, non-OPEC supply surged 2.2 mbd, with most of the supply coming from Americas. US oil producers are leading the surge with US supply rising a massive 1.5 mbd over year ago levels, with ~1 mbd gains coming from crude and condensate alone. After reaching an all-time high in 2022, US oil liquids supply (crude, condensate and natural gas liquids) is on track for a second consecutive record in 2023. Meanwhile, record output from Brazil and Guyana contributed another 600 kbd. With non-OPEC+ supply more than sufficient to fully cover global oil demand, OPEC alliance has been left having to cut production to balance the markets ([The Labors of Sisyphus](#), 14 June 2023). Even with the bloc's effective spare capacity at a massive 4.1 mbd—historic high at a time of record demand—Brent oil price has averaged \$82.62 year-to-date, 17% lower than in 2022 (**Figure 1**).

Figure 1: OPEC and Russia spare capacity

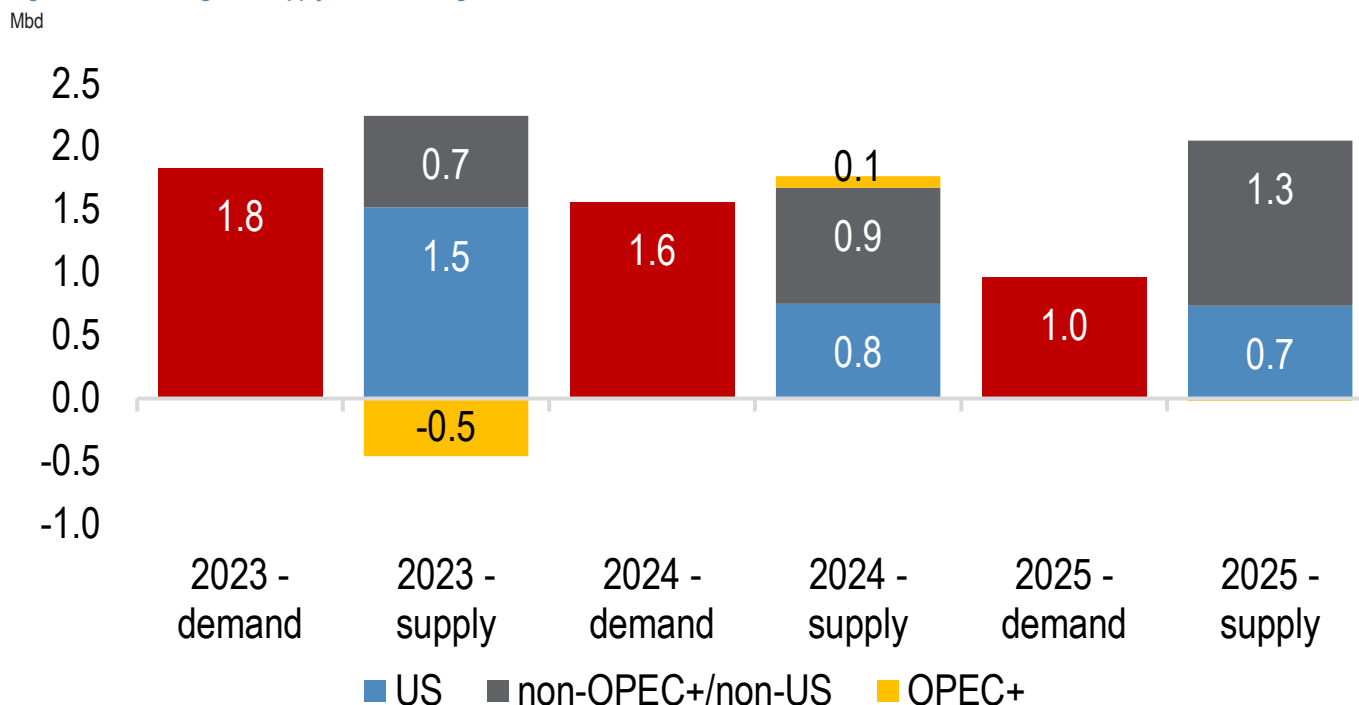
Mbd; The shaded areas represent recessions as defined by the National Bureau of Economic Research (NBER)



Source: S&P Platts

In 2024, we expect more of the same. Despite sustained economic headwinds, we hold a positive view on demand, underpinned by robust EM, resilient US and stable but weak Europe. The world's oil demand is expected to increase by a solid 1.6 mbd in 2024 to a record annual high of 102.8 mbd. Two-thirds of demand gains would come from the overall economic expansion, while one-third of the growth next year will likely be still driven by continued normalization of mobility fuels (mostly jet). Non-OPEC+ producers should again drive overall growth in supply in 2024—projected at 1.7 mbd—more than matching growth in demand (**Figure 2**). US gains will likely slow to 800 kbd, boosting total liquids production to a fresh annual high, and Canadian oil supply should rebound by an estimated 500 kbd after production recovers from fires - and maintenance-depressed 2023 levels. Guyana—where Exxon Mobil has started production at a third offshore oil development in November 2023, brining the total capacity to about 620 kbd—will likely deliver another 170 kbd of annualized growth, with Argentinian oil supply rising 75 kbd. To keep the oil market balanced, OPEC+ alliance would need to continue to constrain production. Essentially, following a 1.3 mbd decline in 2023, oil production by OPEC+ group would have to stay flat in 2024 with Saudi production at a level of what the country produced in 2020 (**Figure 3**).

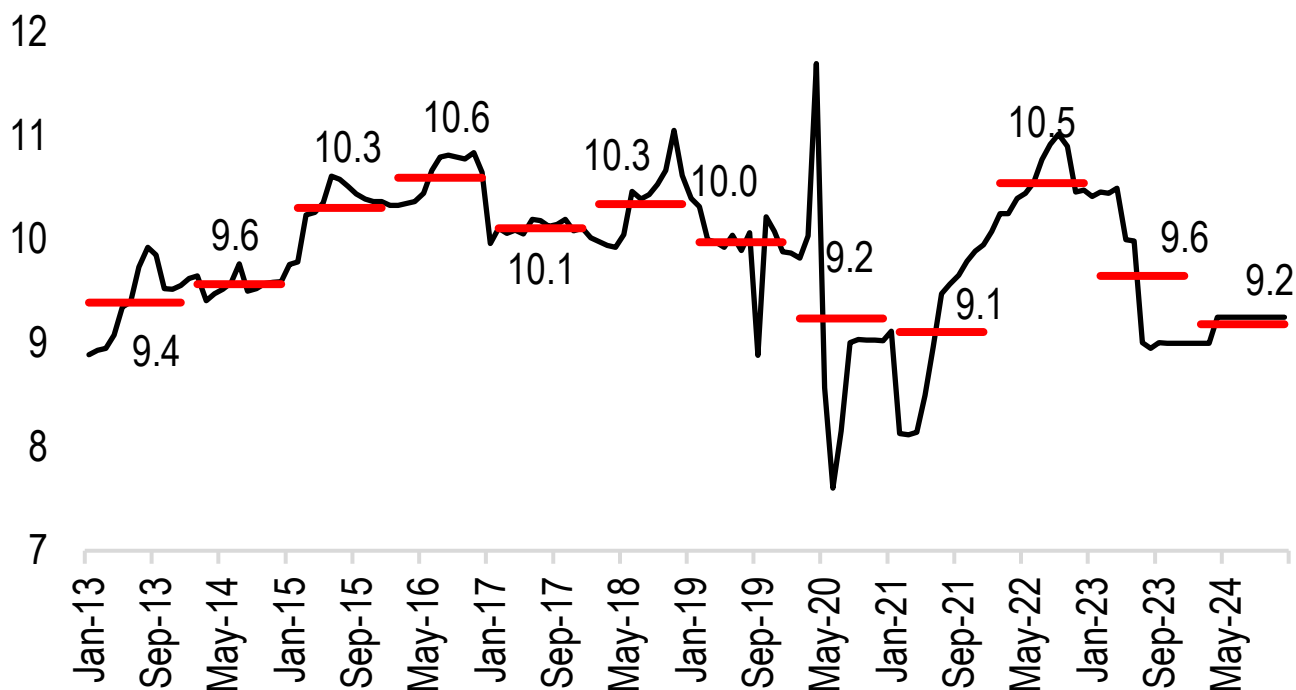
Figure 2: 2023-2025 global supply and demand growth



Source: IEA, Platts, Wood Mackenzie, Rystad, J.P. Morgan Commodities Research

Figure 3: Saudi Arabia crude oil production

Mbd



Source: OPEC, S&P Platts, Rystad, J.P. Morgan Commodities Research

Global oil demand growth is expected to decelerate to 1 mbd in 2025 as the last phase of the post-pandemic rebound dissipates and advancing energy efficiencies and an expanding electric vehicle fleet gain ground. Again, non-OPEC+ supply should have no difficulty keeping up with demand. Record output from the US, Brazil, Canada and Guyana should underpin a 2 mbd increase in oil supplies to a record 105.8 mbd. While the questions hang over the ability of US producers further to increase production in response to demand growth, we believe US is capable of delivering another 700 kbd of growth in 2025, assuming our 2024 price forecast of \$83 Brent materializes. Importantly, with Permian well characteristics trending lighter, half of that gain will come from natural gas liquids. Lower 48 should contribute another 320 kbd, most of that from the Permian, and the Gulf of Mexico will likely grow 85 kbd. The share of US oil in total non-OPEC+ supply growth should decline in 2025 as large-scale offshore developments in Brazil, Guyana, Suriname and Norway continue to ramp up. Reversing a long decline in spending on the decades-long projects, global oil companies are pumping billions of dollars into offshore drilling. While initially expensive and time consuming, offshore production relies on massive economies of scale, producing vast amounts of oil for decades at a time. In total, eight floating production, storage and offloading (FPSO) vessels with combined capacity of 1.4 mbd are committed for delivery in 2025. The commissioning of five FPSOs in Brazil's offshore oil fields should boost the nation's oil output by over 400 kbd in 2025. Meanwhile, the expected delivery of the fourth FPSO in waters off Guyana would bring Exxon's total Guyanese capacity to over 800 kbd by the end of 2025, a major milestone towards reaching a combined capacity of more than 1.2 mbd by year-end 2027. Across the Atlantic, the delivery and ramp up of the Johan Gastberg FPSO should boost Norway's production by 190 kbd in 2025, while Senegal's first offshore oil project will likely begin operating in 2024 and deliver 100 kbd of oil supply in 2025. In Canada, continued growth from oil sands production and shale drilling will likely add 200 kbd of production, while continued investment and development in Argentina should see an additional 80 kbd growth from the nation's shale drilling activity.

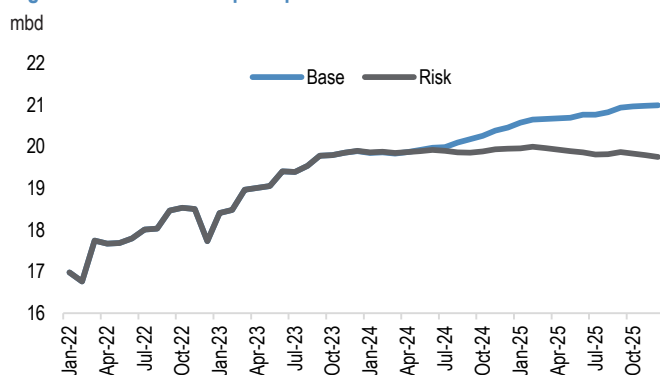
With demand growth set to slow, and non-OPEC supply surging, the market would shift into large surplus, dragging Brent oil price into \$60s by year-end 2025. **Invariably, to make room for this supply growth, OPEC+ members would need to deepen current cuts by 1 mbd, were the alliance to adhere to market management.** (For a differentiated view, see [GES—Supercycle returns. Upgrade Global Energy to OW](#), Malek et al., 21 September 2023).

Bring it on

A great deal will depend on whether Saudi Arabia and Russia continue to voluntarily constrain their production in 2024 and 2025. Our balances and path of the price forecast suggests that as long as OPEC+ members maintain the 2 mbd cut to quotas in place since November 2022 and the extra 1.7 mbd of curbs from some members that started in May 2023, Saudi and Russia will be able to unwind some of their voluntary reductions at the start of 2Q24. Assuming Saudi Arabia pumps additional 250 kbd starting from April and through the end of 2025 and Russia increases its exports by 150 kbd over the same period, global oil inventories would likely build by 100 kbd in 2024 and 1.2 mbd in 2025. In this scenario, Brent oil price should average \$83 in 2024 and \$75 in 2025.

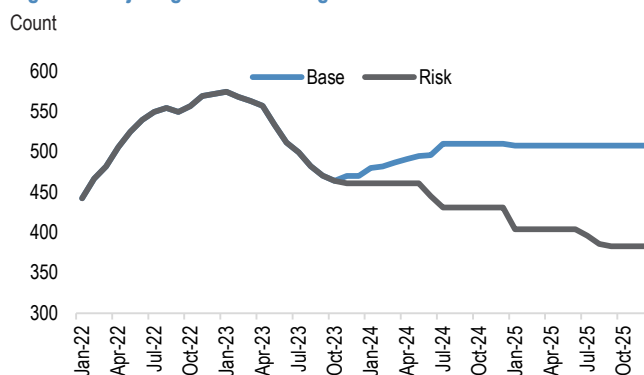
However, a case could be made for choosing higher volumes in order to achieve better optionality in the future. Were Saudi and Russia to bring the full 1.3 mbd of self-imposed curbs into the market in April, global oil surplus would surge to 700 kbd in 2024 and 2.1 mbd in 2025. Our pricing model suggests that the price impact on 2024 would be about \$6/bbl annualized, to \$77. Hit to the 2025 price would be substantially larger, with Brent oil averaging \$57, assuming no response from price-sensitive supply. However, with WTI price dipping below \$70 in 4Q24 under such scenario, US oil operators would likely react. Majority of the US operators have already outlined and begun to execute their 2024 drilling programs, and it would likely take a sharp drop in the price outlook to make them change their strategies. Full unwind of the 1.3 mbd voluntary cuts and the resulting drop in price, however, would likely trigger a slowdown in drilling and fracking activity in the second half of 2024, shaving about 170 kbd off our 760 kbd projected growth in US oil output in 2024. In 2025, US oil and condensate production will likely contract by 150 kbd, while NGL output will continue to grow by 150 kbd, keeping total US liquids production flat compared to our baseline projection of 740 kbd growth (**Figures 4 & 5**). In turn, this should boost 2025 Brent price by \$7 to an average of \$65.

Figure 4: Total US oil liquids production



Source: EIA, Rystad, J.P. Morgan Commodities Research

Figure 5: Major tight oil basins rig count



Source: Baker Hughes, J.P. Morgan Commodities Research

US administration to purchase 15 mn bbl for SPR in 2024

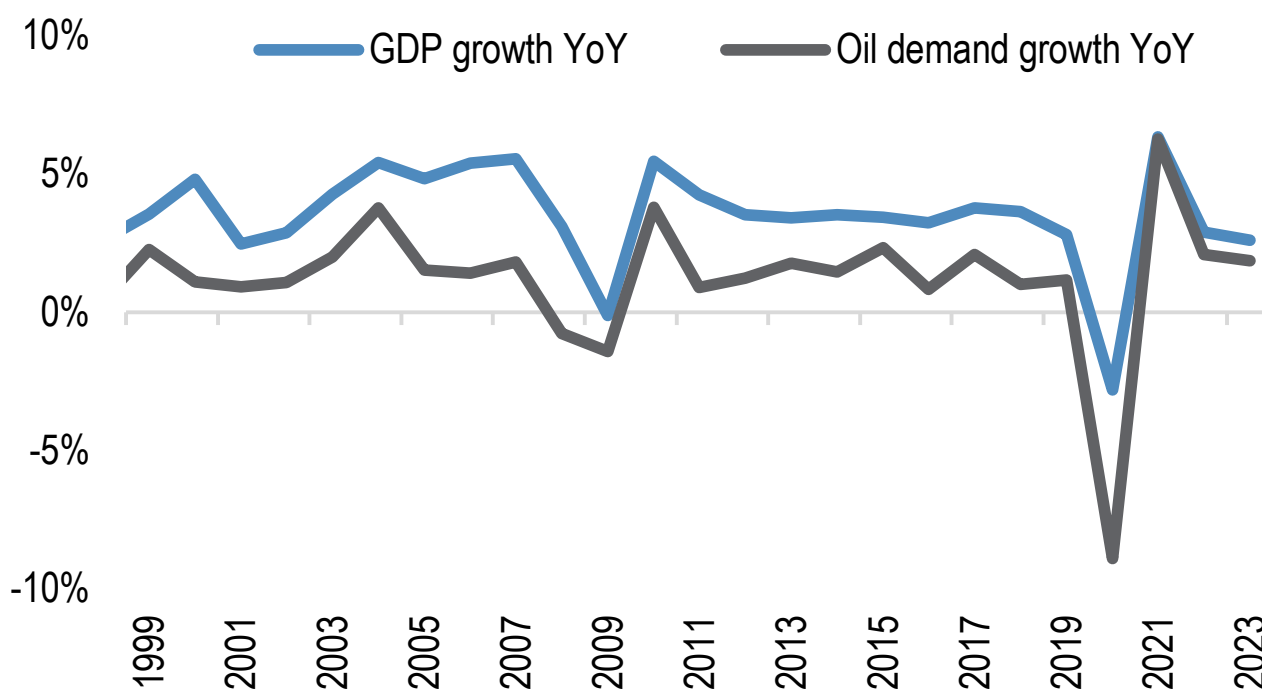
US administration will continue to refill the SPR in 2024 when WTI oil price dips below \$79/bbl. According to our price forecast, we believe the DOE would be able to purchase 15 million barrels between January and July 2024. The buying program should have only a limited impact on prices, given DOE's refill rate of 3 mbd per month or 100 kbd. For comparison, in 2023, despite expressing willingness to refill the nation's oil reserves, the administration has actually purchased only 4.8 million barrels, citing technical constraints for lack of action.

Demand growth to ease to a still solid 1.6 mbd in 2024

Oil demand in the next two years will depend on the health of global economic growth and on the oil intensity of this growth. Key to the call is our view that the the bulk of the world's and China's pent-up demand recovery is likely behind us and that, for the first time since 2020, oil demand will re-link with global economic growth. Because of the nature of restrictions and the oil intensity of the activities that were effectively shut down, lockdowns had an outsized drag on global oil demand in 2020-2022—well below the levels prescribed by the economic output—and an outsized boost when restrictions were fully lifted in 2023 on the back of China's reopening—despite a subpar GDP growth (**Figure 6**). Entering 2023, demand for mobility fuels like gasoline, diesel, and jet fuel remained 2.8 mbd below their pre-COVID levels (gasoline -0.7 mbd, diesel -0.9 mbd, and jet -1.2 mbd). The release of this pent-up demand contributed almost half of the 1.9 mbd global demand gain in 2023. Today, the world's diesel and gasoline consumption are already on par with pre-pandemic levels, while petrochemicals demand surpassed 2019 levels by almost 800 kbd. Only jet fuel remains ~800 kbd below pre-pandemic volumes, leaving room for continued normalization.

Figure 6: Global GDP and oil demand growth

Percent YoY



Source: IEA, IMF, J.P Morgan Commodities Research

From here, we expect global oil demand growth to be slower and more aligned with GDP, which we project to shift lower from 2.6% in 2023 to a below-potential 2.2% pace in 2024, reflecting the delayed impact of tighter monetary policy in advanced economies. Accordingly, after rising 1.9 mbd in 2023, we expect oil consumption growth to slow to 1.6 mbd in 2024.

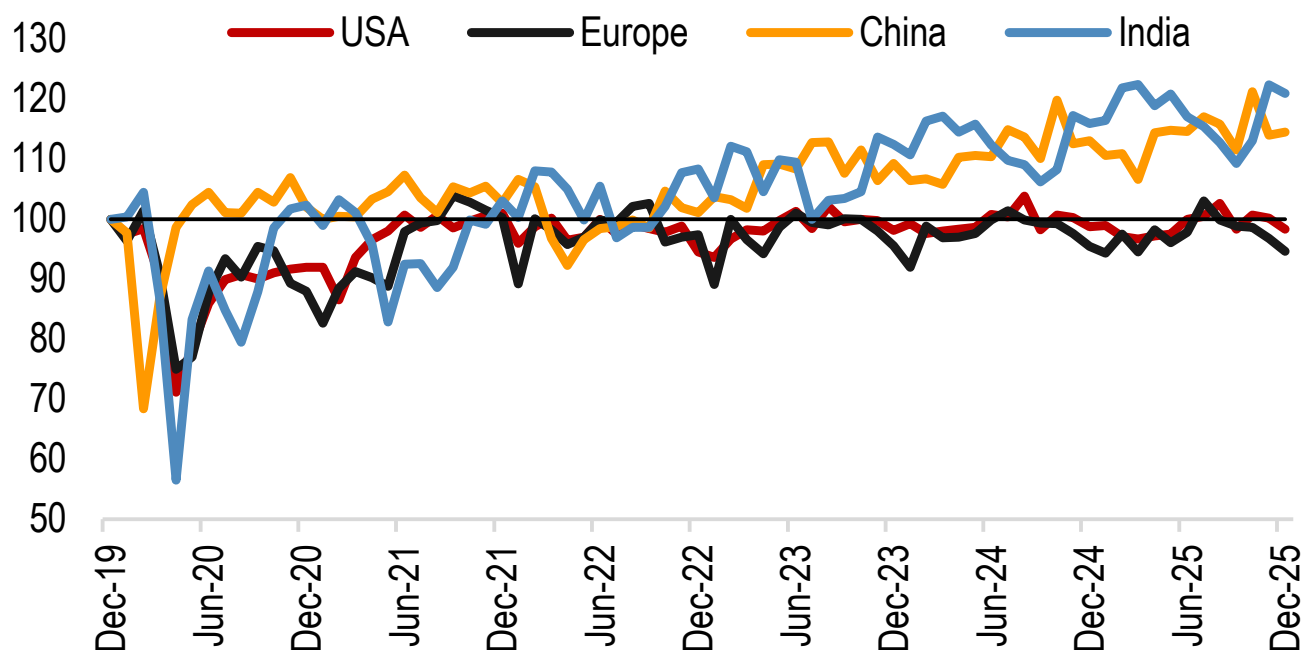
Overall, our outlook on oil demand in 2024 remains positive. Absent a deep recession, the long-run demand elasticity of 0.46 at the projected 2.2% GDP growth rate implies an optimal oil demand growth rate of about 1.1 mbd in 2024. In addition to oil demand driven purely by global economic expansion, we see additional 0.5 mbd of demand

coming from the continued normalization in air travel and APAC petchem demand. Although air travel and tourism will continue to recover throughout 2024, we assume only half of the remaining 800 kbd of pent-up jet fuel will be consumed next year and see jet demand surpassing 2019 levels only in May 2025. This is mostly due to multiple policy changes targeting the use of jet and kerosene in residential, commercial and agricultural sectors. For example, Japan, which accounts for 260 kbd of the global 830 kbd of non-aviation use of jet/kero, has seen demand drop to 200 kbd in 2023 due to ongoing efficiency improvements and energy transition efforts. Similarly, India's kerosene use has declined by nearly a quarter since 2019 due to the government's efforts to switch to cleaner alternatives like LPG. We expect further declines in India's jet/kerosene usage in 2025.

Regionally, out of the 1.9 mbd rise in demand this year, China, India and other Asian countries contributed 1.5 mbd, or 80% of the growth, with Middle East, Africa and Latin America adding another 140 kbd. Oil demand in the US and Europe rose a combined 160 kbd, while demand contracted slightly in Japan and South Korea. In 2024, Asia again will dominate oil demand growth, with China (+500 kbd) and other China-exposed economies adding up to about 910 kbd, or 60%, of the rise in global consumption. Oil demand will remain tepid in the developed world, with low-positive increases in the US (+160 kbd), Europe (+80 kbd), and South Korea (+110 kbd) and contracting in Japan (-50 kbd) (Figure 7).

Figure 7: Regional total oil demand

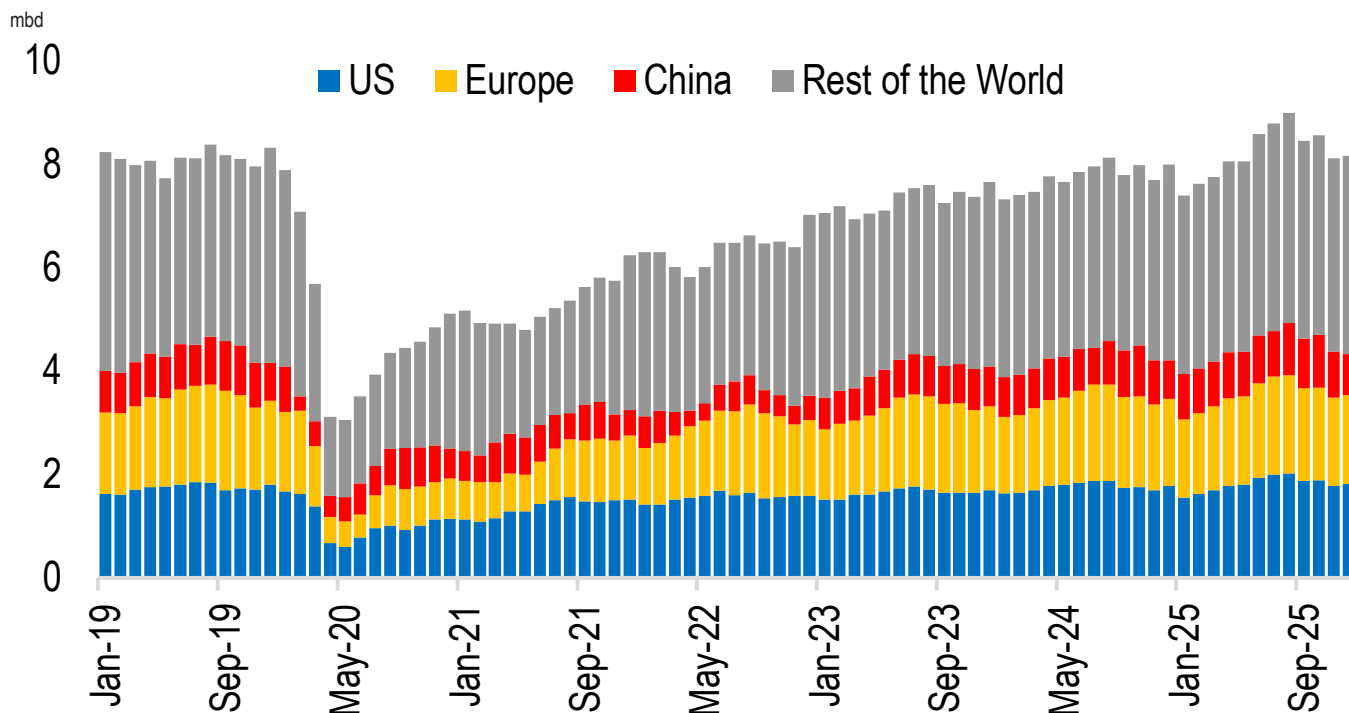
Index, December 2019 = 100



Source: Wood Mackenzie, IEA, FlightAware, EIA, MarITrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan Commodities Research

By product, jet fuel consumption is seen rising 450 kbd to 7.8 mbd in 2024, putting jet demand just 4%-pts below pre-pandemic levels. We estimate jet fuel demand will end this year at 94.3% of 2019 levels and climb further to 100.8% by December 2024. The US, Europe and China will account for lion's share of this growth, as international travel between these regions continues to normalize. The Civil Aviation Administration of China and the US Department of Transportation announced at the end of October that the number of flights will increase from the current 48 (24 round trips) a week to 70, starting November 9 (Figure 8).

Figure 8: Global jet fuel demand by region



Source: Flightaware, IEA, EIA, Wood Mackenzie, J.P. Morgan Commodities Research

World's gasoline demand likely surged 750 kbd this year, with China accounting for half of that growth as road traffic rebounded sharply after China reopened. Our forecast calls for global gasoline demand growth to ease substantially to 170 kbd in 2024 to an average of 26.9 mbd, 100 kbd above 2019 volumes. Emerging markets—boosted by India, APAC and Africa—will likely see their gasoline demand rising 270 kbd while developed markets should experience a 140 kbd contraction—the sharpest decline since 2012 outside of COVID (**Figure 9**).

Figure 9: Developed market's gasoline demand change year over year

kbd

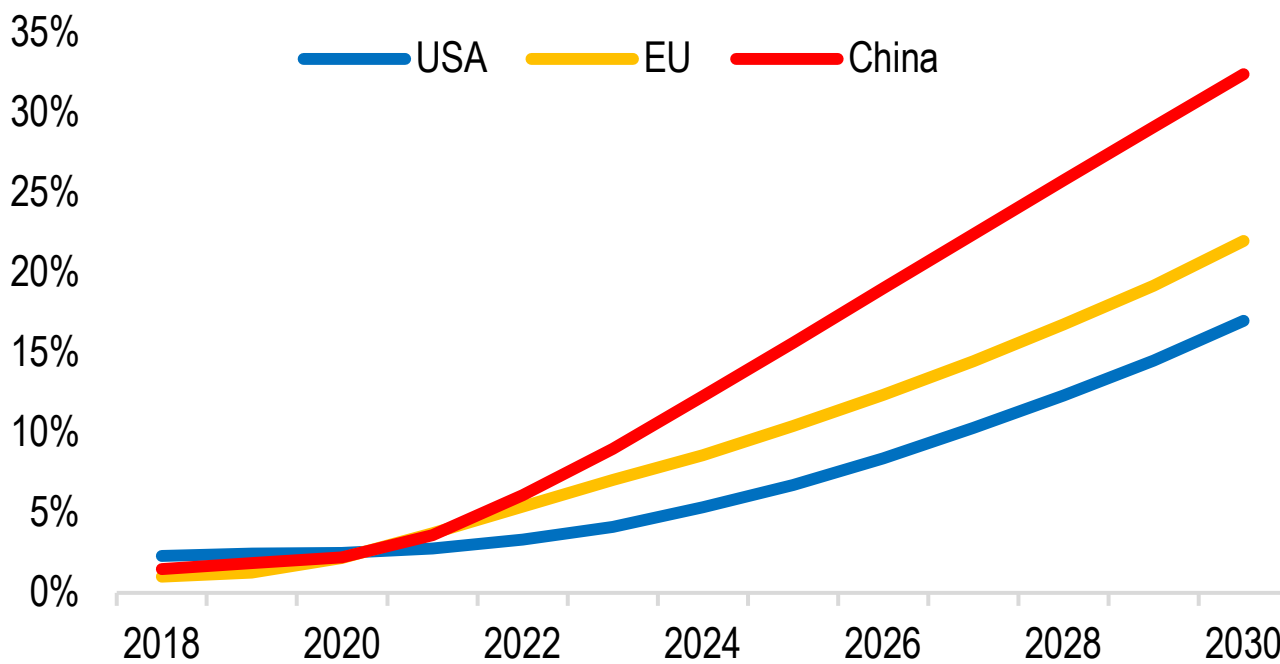


Source: IEA, J.P. Morgan Commodities Research

To put things in perspective, 27% of all cars sold in the world in September were vehicles that run on full or partial electricity and hybrids (so-called xEVs): 18% in the US, 40% in China and 47% in Europe. With xEV sales taking off rapidly, they now comprise 4.1% of the US fleet of about 252 million passenger cars, likely resulting in 150-200 kbd permanent demand loss. In terms of fleet composition in the rest of the world, the xEV segment makes up about 9% of China's 265 million fleet and 7.3% of 274 million light-vehicles on the European roads (**Figure 10**). Consequently, we estimate US gasoline demand will likely contract 70 kbd in 2024 with the decline expected to deepen to 210 kbd in 2025. Europe, Japan and South Korea should also witness a combined 70 kbd contraction in gasoline demand in 2024. China will likely join the club of developed nations with plateauing gasoline demand in 2024.

Figure 10: Share of xEVs (hybrid, plug-in, battery) as a % of total LDV fleet

Percent



Source: US Department of Transportation, Eurostat, Company reports, J.P. Morgan Commodities Research

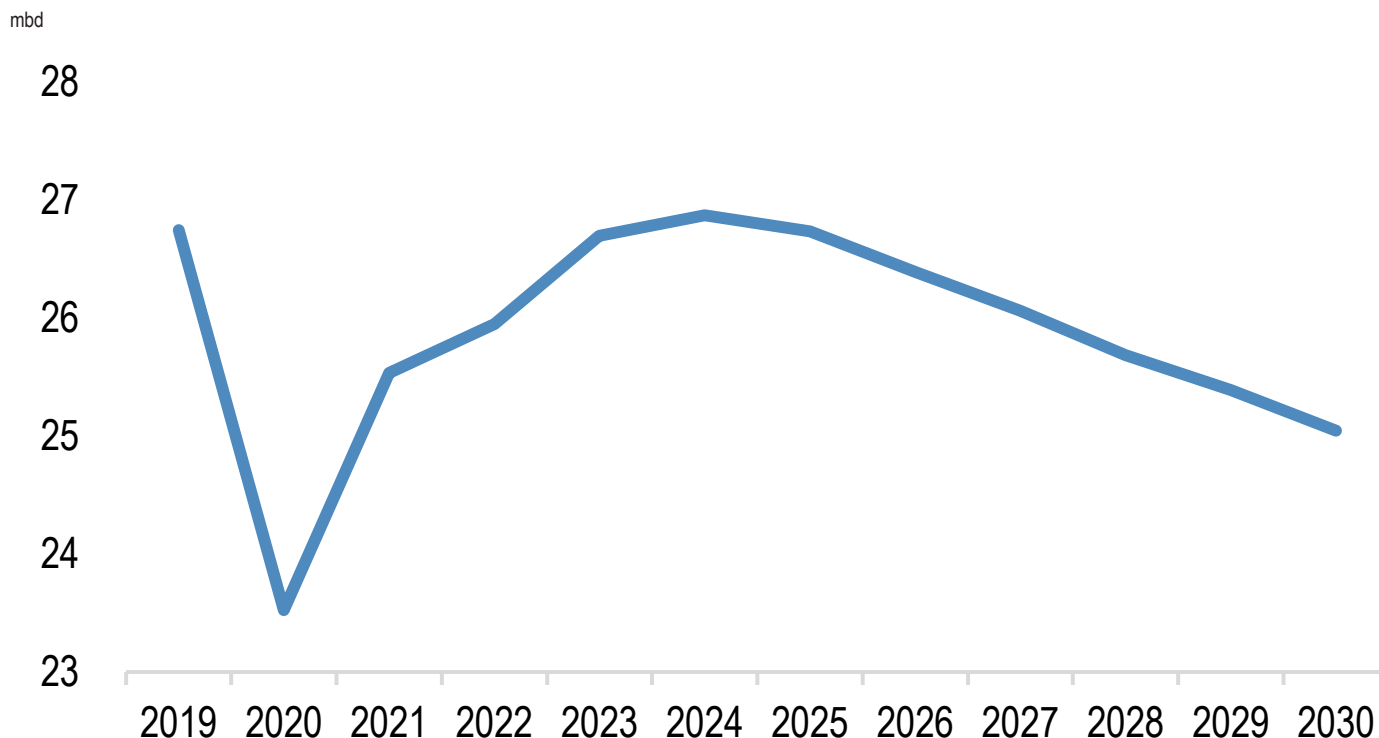
Diesel demand is set to contract marginally (-20 kbd) in 2023, as a 100 kbd rise in industrial use is outweighed by a 120 kbd decline in road applications. We project a 340 kbd recovery in global diesel consumption next year largely boosted by a continued recovery in industrial applications and expansion of the trucking fleet in major emerging economies like India and China. India reported 28% increase in commercial vehicles sales in the 2023 fiscal year that ended in March to 880K and sales are up 8% this year through October. China's commercial vehicles sales are expected to jump 13% in 2023 to 3.7 million vehicles with sales poised to average 3.6 million vehicles each year through 2030.

Global petchem demand is estimated to rise by nearly 300 kbd in 2024 with nearly 200 kbd of that increase coming from China, where 700 kbd of new refining capacity coming on stream in 2024 will require ever higher volumes of feedstock. With China shifting away from imports of petchem products to domestic production in line with the objectives outlined in the current 14th Five-Year Plan, it began to greatly expand its ethylene capacity. China's imports of feedstock like LPG, ethane and naphtha surged in recent years, emerging as a key driver of crude oil demand growth. The corollary of this expansion has been a sharp drop (120 kbd) in naphtha consumption in the neighboring Japan, South Korea and Taiwan—countries that were traditional sellers of petchem products to China. We expect 100 kbd of that naphtha demand to recover in 2H24.

Looking into 2025, we project oil demand growth to decelerate to around 1 mbd to a record 103.9 mbd. After the soft patch in 2024, global economic growth should revert to its long-run trend in 2025, however, decarbonization policies put into place more than a decade ago will begin to cut into demand for some oil products on annualized basis ([Oil 2030: Long Term Incentive Price](#), 28 June 2023). World's gasoline use will likely contract 140 kbd in 2025, with declines set to expand in the coming years (**Figure 11**). Based on guidance from auto makers, by 2025, share of hybrid, battery electric and plug-in vehicles is projected to increase to 24%, 50% and 52% in the US, Europe and China, respectively. The penetration of the xEV vehicles in the fleet will likely also rise to 7% in the US, 10% in

Europe and 16% in China.

Figure 11: Global gasoline demand



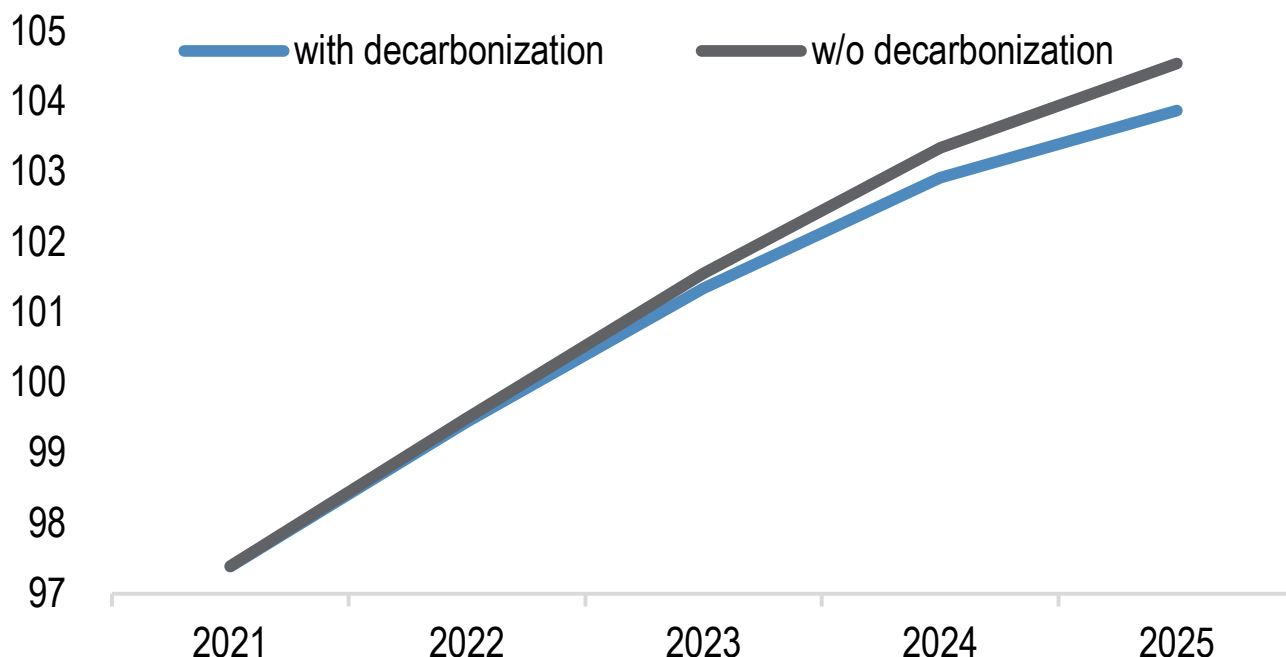
Source: EIA, IEA, Wood Mackenzie, J.P. Morgan Commodities Research

Although large-scale substitution options remain limited in the 29 mbd global diesel market, this is the only other oil product where decarbonization technologies exist on a small-to-medium scale. In the US, we expect the consumption of renewable and biodiesel to increase by 115 kbd by 2025, most of which will directly replace petroleum-based diesel. This will increase the share of biodiesel in US diesel consumption from 7% today to 10% in 2025. Unlike the US, China is moving ahead with the electrification of its heavy-duty vehicle sector, as well as shifting the freight transport from trucks to rail and waterways. Since the announcement of the Clean Diesel Action Plan in January 2019, today, more than 80% of China's diesel-consuming bus fleet has been replaced with electric buses, nearly 477,000 light electric commercial vehicles have been introduced to the vehicle fleet, and rail freight volumes have risen 35% above 2017 levels. We are also observing a steady increase in sales of electric trucks. In September 2023, electric heavy and medium truck sales accounted for 5.3% of total truck sales, up from 1.4% in April 2021. During the same month, sales of electric light trucks jumped to 12.1%, up from 2.9% in April 2021. We project the share of electric commercial vehicles in China to increase from 10% currently to 16% in 2025 and 34% by the end of the decade.

In the absence of supportive electrification policies and fuel efficiency mandates globally, we estimate that global oil demand would have likely averaged higher by 200 kbd in 2023, 400 kbd in 2024 and 700 kbd in 2025 with total global oil demand at 104.5 mbd in 2025 against our current estimate of 103.9 mbd (**Figure 12**).

Figure 12: Global oil demand with and without decarbonization efforts

mbd



Source: Wood Mackenzie, IEA, FlightAware, EIA, MarITrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan Commodities Research

Crucially, although we don't see peak oil demand on the horizon, demand growth should be increasingly driven by petrochemicals and air travel at the expense of traditional oil products like gasoline, diesel and fuel oil. Accordingly, the share of refined oil products will likely account for 88.4 mbd or 85% of oil demand in 2025, with the balance coming from natural gas liquids and biofuels.

Non-OPEC+ supply growth to decelerate to 1.7 mbd in 2024

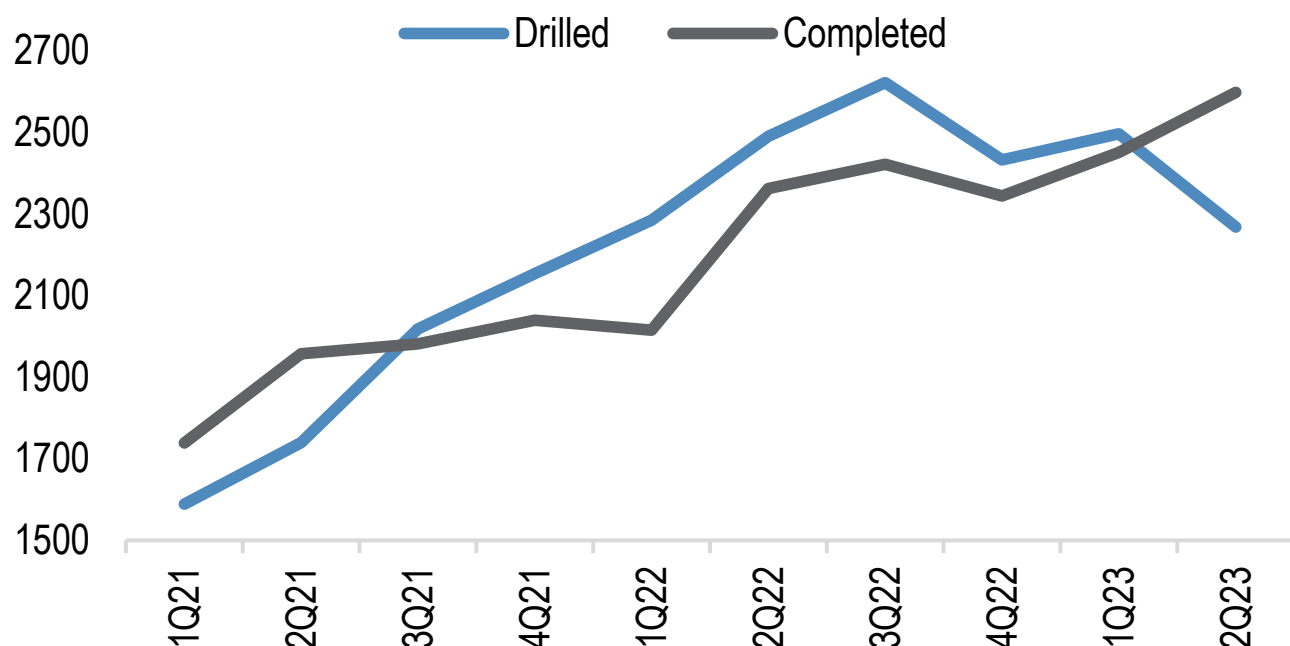
US: US total oil liquids (crude, condensate and NGLs) production is set to rise by 1.5 mbd in 2023, followed by 0.8 mbd in 2024 and 0.7 mbd in 2025. After record 2022, total US oil production is on pace to set another all-time high in 2023. Even though last year's record output was driven by natural gas liquids—NGLs have been posting annualized growth since 2010—this year, supply growth from both NGLs and crude and condensate will contribute to record production. For the eight months of 2023, US crude and condensate production has averaged 12.8 mbd, topping previous *annual* record—set in 2019 just before the pandemic impacted production—of 12.3 mbd. The country remains a net liquids exporter, shipping 3 to 5 million barrels per day of liquids.

Total US crude and condensate production will likely rise 11% this year, more than double the mid-single digit target guided by the public operators at the start of the year. Supply surge is explained by the price of oil, higher completions activity from private operators and technological innovations.

As oil prices surged in the first half of 2022 in the immediate aftermath of Russia's invasion of Ukraine, private operations have rapidly scaled up their drilling activity. But given the standard cycle time of multi-well pad developments, majority of the wells that were drilled in the 2H22 were completed only at the start of 2023. For example, in the tight oil basins, the number of wells drilled peaked in 3Q22, whereas the total number of completed wells peaked likely only in 2Q23, driving production higher (**Figure 13**).

Figure 13: Major tight oil basins drilled and completed wells

Count



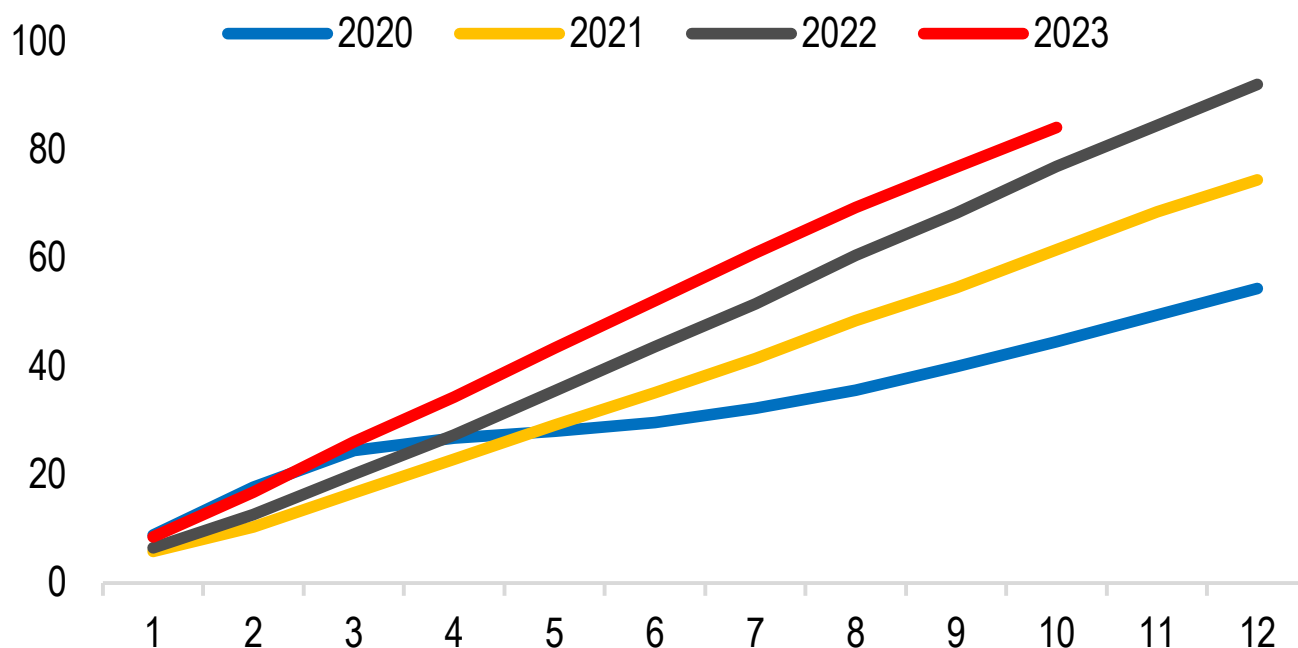
Source: Rystad

Yet, while oil output is growing to record highs, the number of rigs in operation at US fields has declined this year as private operators retreated amid weaker pricing environment and consolidation. In the week to November 10, total rig count fell 163 to 616, the lowest since February 2022. US producers, however, are squeezing more oil out of shale fields via contiguous property leases, longer horizontal drills and more simultaneous fracking. The median lateral length drilled in the major US basins will continue to grow longer with one out five wells drilled this year in the oil-rich Permian basin extending beyond 12,500 feet. Only 1% did so five years ago.

In the completions process, frac crews are expected to outpace 2022 in every measurable metric, including cumulative lateral length stimulated, proppant pumped and number of wells stimulated. Comparing performance over the first 10 months of 2023 vs 2022, the number of active frac fleets in the major tight oil basins has declined 2%, however cumulative length of lateral footage stimulated has increased 8% (**Figure 14**). Accordingly, frac crews have been able to complete a cumulative 6 million more feet of lateral length through October of this year compared to the same period last year, which we attribute to longer laterals and fracturing techniques like simul-frac and the transition of E-fleets.

Figure 14: Cumulative lateral length completed in the major tight oil basins

Million feet

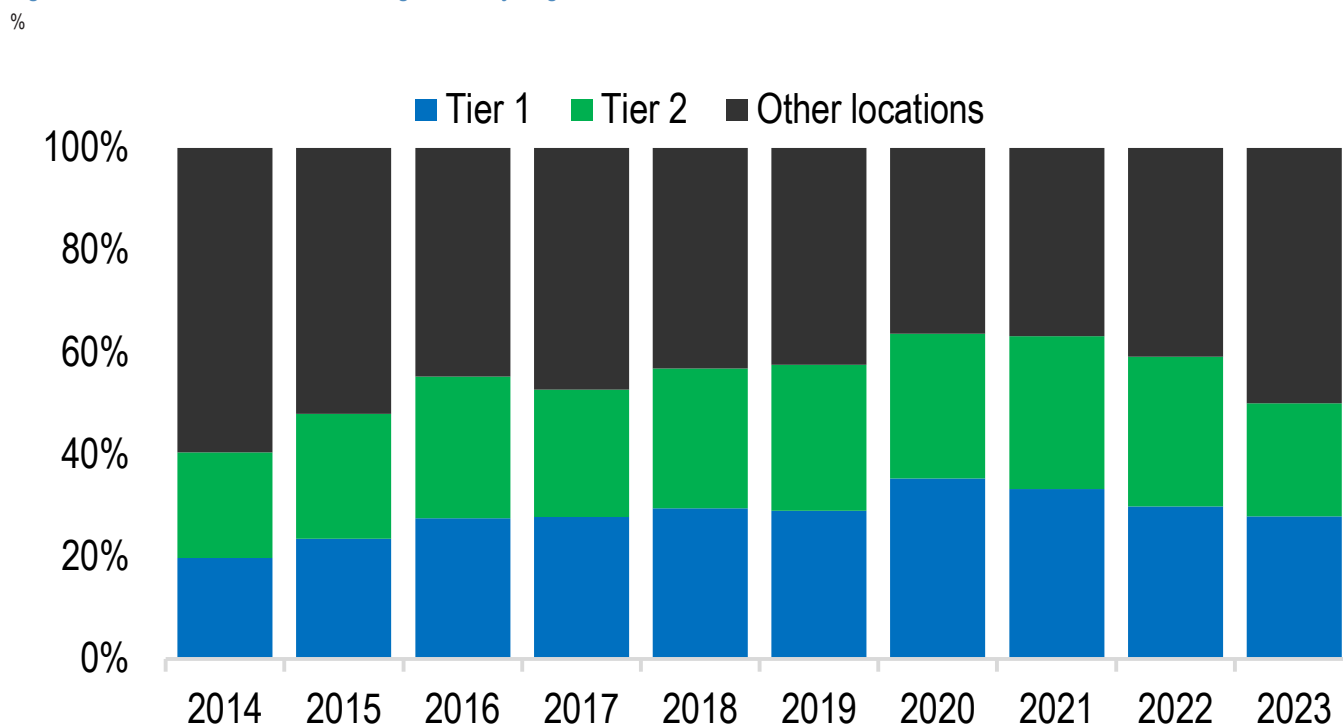


Source: Rystad

In 2024, US crude and condensate production is set to continue to grow, albeit at a slower pace. Our forecast calls for tight oil production to grow by 5%, in line with the low-to-mid-single digit growth guided by public oil and gas companies. Output from the Lower 48 will likely increase by 364 kbd, while Gulf of Mexico adds only 45 kbd and production in Alaska declines marginally (-11 kbd). Our 2024 WTI price forecast of \$79/bbl suggests an oil price that sits comfortably above the breakeven costs for most drilling locations across the US shale basins. We believe this should encourage a recovery in drilling activity, with 40 rigs projected to be added in the major tight oil basins by the end of 2024, three-quarters of them in the Permian. Moreover, following two years of substantial cost inflation, public operators are now guiding for a 5% cost deflation in 2024, which should provide some cost relief in boring new wells. We note that the rig additions will not have any significant impact on production until 2025 due to the time lag it takes for drilled rigs to filter through a change in output.

The share of drilling and completions activity on tier one acreage has been trending downward since 2020, declining on average by about 2% per year, and will likely continue to decline in 2024, requiring more wells to be drilled in order to offset the shift in the well performance (**Figure 15**). In general, lower tier acreage suffers from weaker well performance, which in turn results in higher breakeven costs. However, observed efficiencies on the completions side of the business have so far helped to offset some of the drop in individual well performance.

Figure 15: Share of tier 1 and tier 2 drilling in the major tight oil basins



Source: Rystad, J.P. Morgan Commodities Research

Permian should continue to lead US oil production growth in 2024, accounting for 82% of the incremental shale supply. Outside of the Permian, oil production in Anadarko, Bakken, Eagle Ford and Niobrara basins is set to grow only a marginal 75 kbd in 2024 compared to 240 kbd increase in 2023. Following a 150 kbd jump in production in 2023, crude and condensate output in the Gulf of Mexico will likely rise only 45 kbd in 2024, as Mad Dog Phase II, Vito and King's Quay operations ramp up. Oil production in the Gulf has been rising on year-over-year basis since 2021 and is expected to continue to grow as operators approve new projects and maintain capital investment in the region.

Total US oil supply is set to increase another 740 kbd in 2025, with crude and condensate supply rising 410 kbd. Many of the same forces will be in play in the Lower 48, whereas the Gulf of Mexico will get an additional boost in production due to seven new projects coming online. We believe the risk to our oil supply forecast is skewed to the upside given our view on natural gas prices ([2024's bearish uncertainty lends to 2025's bullish certainty](#), 22 September 2023). With multiple LNG trains set to begin loading in the second half of 2024 and into 2025, US natural gas prices are expected to rise above the current forward curve, at \$4/MBtu by 2025, reducing the breakeven costs of drilling oil wells. Higher price for associated natural gas will help to unlock additional drilling locations, likely increasing drilling and completion activity across the major tight oil basins.

NGLs to grow by 363 kbd in 2024 and an additional 326 kbd in 2025. US natural gas liquids production has continued to play a crucial part in the US's total liquids production as output has now reached 6.5 mbd in August – a new record according to the EIA. Since the start of the shale boom in 2014, NGL production has never reported a yoy contraction in production, a trend that we do not expect to change (**Figure 16**). As shale operators continue to expand their drilling programs further out from the core, increasingly widening gas-to-oil ratios (GOR) support additional liquids extraction from the gas stream.

Figure 16: Historical and forecasted total US liquids production growth

Year	Crude and Condensate (mbd)	yoy (kbd)	NGLs (mbd)	yoy (kbd)
2014	8.8	1294	3.0	408
2015	9.4	653	3.3	329
2016	8.8	-595	3.5	167
2017	9.4	511	3.8	273
2018	10.9	1592	4.4	586
2019	12.3	1362	4.8	457
2020	11.3	-988	5.2	350
2021	11.3	-66	5.4	243
2022	11.9	649	5.9	514
2023E	12.9	972	6.4	494
2024E	13.3	396	6.8	363
2025E	13.7	411	7.1	326

Source: EIA, Rystad, J.P. Morgan Commodities Research

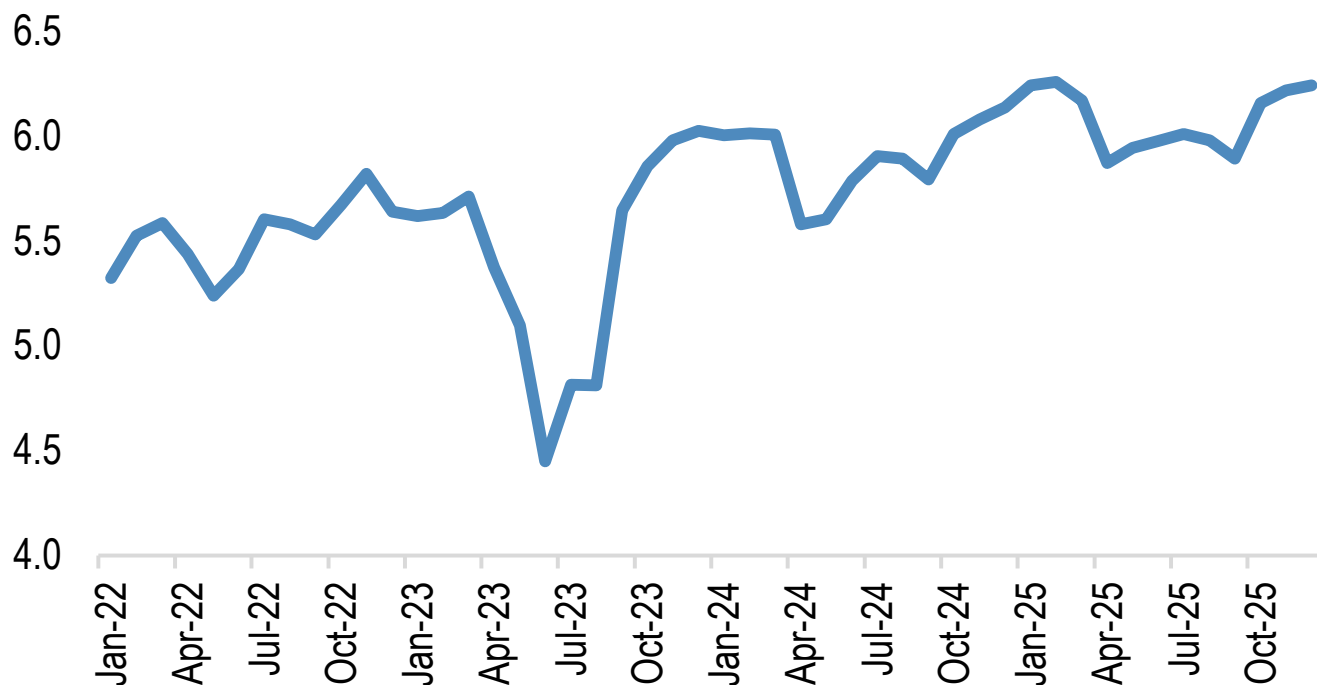
Canada: oil production is expected to increase from 5.4 mbd in 2023 to an annual average of 5.9 mbd in 2024 (+480 kbd), and 6.1 mbd in 2025 (+180 kbd). Following a year headlined by wildfires and heavy maintenance cycle in the oil sands, total oil liquids production in Canada is expected to rebound by 480 kbd in 2024. The flattering base effect is part of this strong growth in supply: the wildfires that began in March and increasingly intensified starting in June had a significant impact on wellhead production in Alberta and parts of British Columbia, likely resulting in a 110 kbd contraction in the 2023 liquids output. Next year, we expect to see a moderate increase in drilling and completions activity across the core plays of Cardium, Charlie Lake, Duvernay, Mannville, Montney and Spirit River, but well productivity gains as well as optimization of production levels at oil sands assets should boost total liquids output to 5.9 mbd, up 480 kbd YoY.

A more structural driver behind our projections of strong growth in Canadian oil output in 2024 and beyond is the approval of incremental pipeline capacity. Takeaway capacity constraints to export markets have long plagued WCSB plays but as the Coastal Gas Link pipeline starts loading in 2024—which will feed LNG Canada’s phase 1 project starting in 2025—and Trans Mountain Express (TMX) begins shipping crude in 2Q24, it will help improve optionality for operators and ultimately increase rig count and production. The 600 kbd TMX pipeline is primed to carry heavier oils and will be utilized by oil sands producers to carry western Canadian crude to the west coast (tidewater terminal in Burrard Inlet) and then to Asia. The Coastal Gas Link pipeline will allow for extra takeaway capacity from the shale basins, given operators the opportunity to increase both their natural gas and associated liquids volumes.

Canadian offshore shelf production is also set to get a slight boost from the restart of the Terra Nova field off the coast of Newfoundland and Labrador. Since the FPSO was pulled from production in 2019 to undergo substantial repairs, the re-start date has continually been pushed back due to low commodity prices and extended work. The joint venture between Suncor, Murphy and Cenovus could see an additional 30 kbd of production return to the region in 2024 as the FPSO was sent out for delivery back in August 2023 (Figure 17).

Figure 17: Canada's crude, condensate and NGL production forecast

mbd



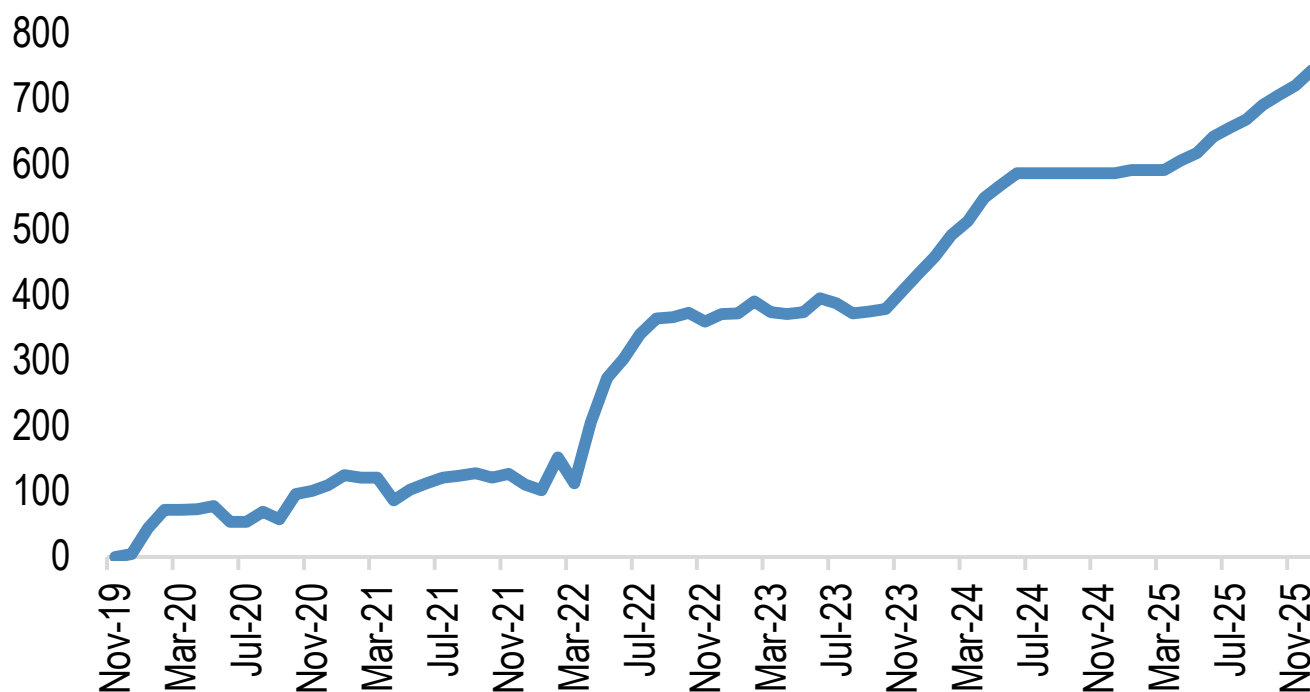
Source: Rystad, J.P. Morgan Commodities Research

Guyana: oil production to increase from 386 kbd in 2023, to 557 kbd in 2024 and 650 kbd in 2025. Guyana's oil production continues to enjoy the fastest growth rate among oil producing countries, surging from zero in 2019 to 386 kbd in 2023. The two FPSOs operating in the region—Liza 1.0 Destiny and Liza 2.0 Unity—have a combined nameplate capacity of 340 kbd, although production from the two platforms has been averaging close to 400 kbd due to debottlenecking and optimization. In 2024, annualized production is expected to increase by another 160 kbd as Prosperity FPSO comes online in the Payara field. With a nameplate capacity of 220 kbd, the Prosperity platform was officially delivered to Guyana in April 2023 and is expected to slowly increase loading volumes starting in mid-November. Total production capacity of the Prosperity FPSO will be likely achieved during 2Q24, boosting total Guyanese oil production above 600 kbd sometime in 2H24.

In line with ExxonMobil and now Chevron's plans to reach 1.2 mbd of delivered nameplate capacity from the region by 2027, the fourth FPSO One Guyana (250 kbd) will be commissioned in the Yellowtail field in 2025, lifting production to an annual average of 650 kbd (**Figure 18**).

Figure 18: Guyana crude oil production forecast

kbd



Source: Rystad, J.P. Morgan Commodities Research

Brazil: crude and condensate production to average 3.3 mbd in 2023, increase further to 3.5 mbd in 2024 and reach an annual average of 3.9 mbd in 2025. Following a period of stagnated oil production in 2020-2022, Brazil's total oil liquids output will likely increase almost 400 kbd in 2023, and rise another 180 kbd in 2024. Led by the state-owned Petrobras, high levels of supply additions stem from both pre-salt and post-salt oil projects. In Brazil's pre-salt layer, Petrobras's current five-year plan calls for an additional 10 FPSOs to be delivered by 2027, with deliveries of FPSO Sepetiba and FPSO Marechal Duque de Caxias set to drive production growth in 2024. The FPSO Sepetiba was delivered to the Mero field in September 2023 and is expected to begin ramping up its 180 kbd capacity by the end of the year. The second platform, the FPSO Marechal Duque de Caxias, is part of the third phase of the Mero field development with a production capacity of 180 kbd and is set for delivery sometime in 2024.

In Brazil's post-salt layer, a combination of FPSO deliveries, the restart of Equinor's Peregrino field and increased maintenance spending from Petrobras have helped to reverse production declines in the region, which now contributes to Brazil's top-line oil production growth.

Looking beyond 2024, current production plans from Petrobras call for an additional 865 kbd of capacity to be added in 2025 through the commissioning of the P-78, P-79, Maria Quitéria, Almirante Tamandaré and the Alexandre de Gusmão FPSOs. Given the timing of the commissioning and the time needed to ramp up to capacity but also production declines from other platforms, oil production from these five offshore platforms would likely annualize to only 405 kbd of growth in 2025 (Figure 19).

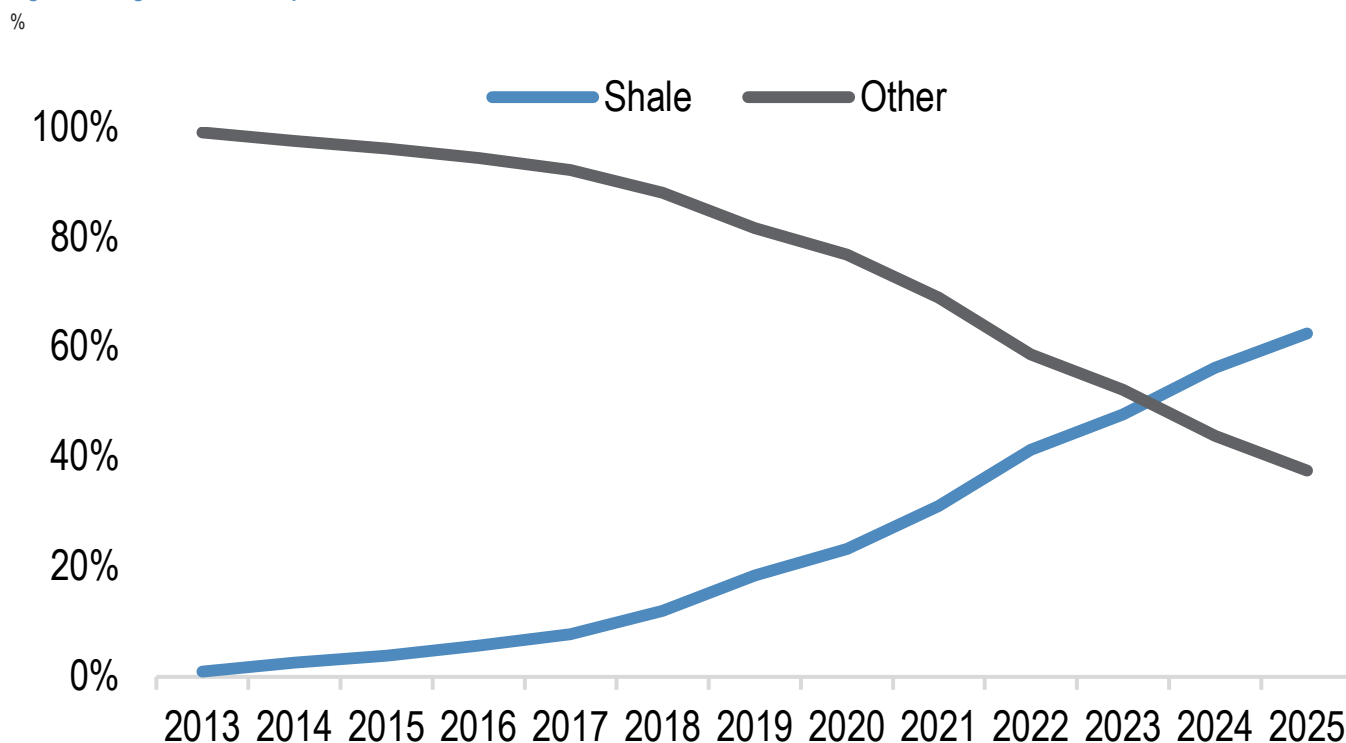
Figure 19: Brazil FPSO delivery schedule 2023-2025

Unit	Start-Up	Capacity (kbd)
FPSO Sepetiba	2023	180
FPSO Marechal Duque de Caxias	2024	180
FPSO Maria Quitéria	2025	100
FPSO Almirante Tamandaré	2025	225
P-78	2025	180
P-79	2025	180
FPSO Alexandre de Gusmão	2025	180
Sum		1,225

Source: Petrobras company reporting

Argentina: crude and condensate production to average 690 kbd in 2023, 770 kbd in 2024 and reach an annual average of 850 kbd in 2025. While Argentina's state-owned oil and gas company YPF has led the development at Vaca Muerta shale play, global majors like Chevron, ExxonMobil, Shell and Total Energies have brought in investment and expertise in unconventional oil and gas extraction. Argentina's oil production rose from 617 kbd in 2021 to 738 kbd in 2023. Vaca's superior oil yield per foot in horizontal wells—Vaca Muerta's breakeven costs sit today at around \$37/bbl—boosted the share of unconventional oil volumes in Argentina's overall crude oil production from 4% in 2015 to 40% last year and is expected to contribute almost half of total oil production this year (**Figure 20**). One of the major key drivers to unlocking oil and gas production from the Neuquen Basin shale formation is the development of takeaway pipeline capacity, which the government has recognized and worked hard to develop. The recent commissioning of the 270 kbd Nestor Kirchner gas pipeline and the plans to double its existing capacity is a testament of supportive government policies. In 2023, capital spend in the basin will likely increase 10%, leading to a 75 kbd growth in oil liquids production in 2024 and another 83 kbd in 2025. There is limited risk to our view on Argentina's oil production from the upcoming presidential elections on November 19, in our view. While the rival parties have contrasting strategies to develop the shale play, the overall objective of both factions is to support investment and development of the asset.

Figure 20: Argentina share of production source



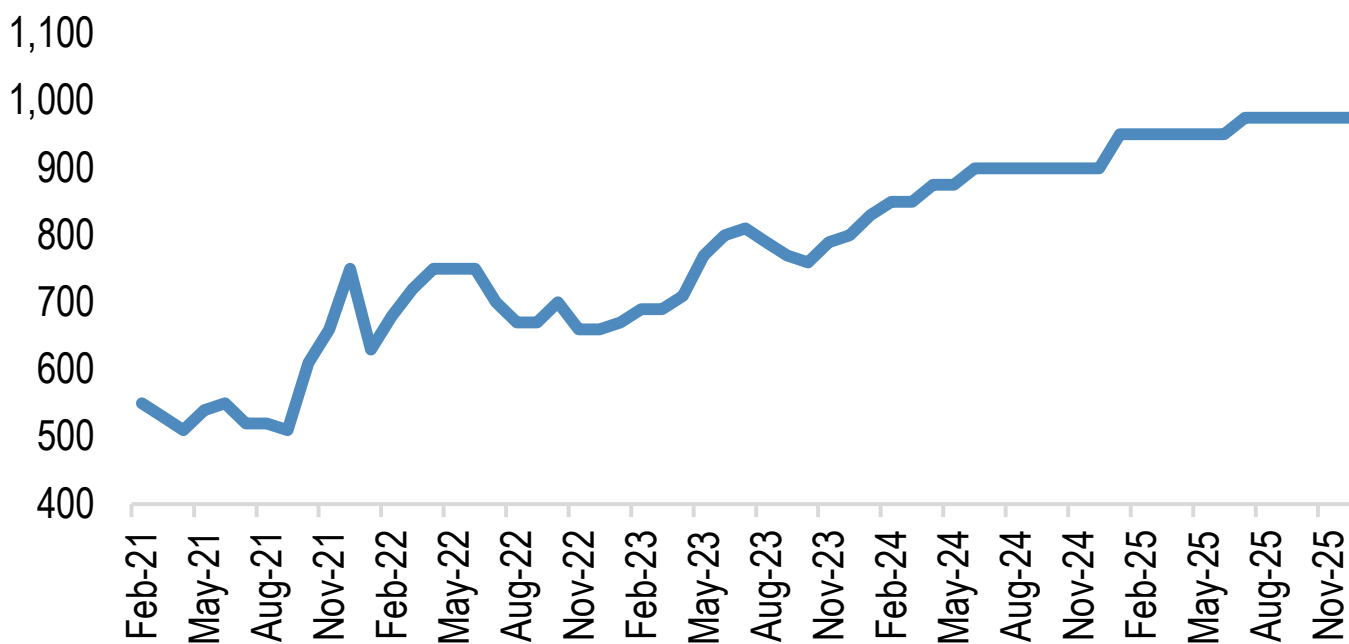
Source: Rystad, J.P. Morgan Commodities research

Venezuela: the lifting of sanctions on PDVSA is expected to gradually increase Venezuelan oil output from 760 kbd in 2023 to 880 kbd in 2024 and 963 mbd in 2025. Venezuelan oil supply already grew 110 kbd between December 2022 and September 2023 to 770 kbd, following the provision of limited waivers to Eni, Repsol and Chevron. The new temporary [sanctions relief](#) introduced in October 2023 effectively lifts all restrictions on PDVSA crude exports and diluent imports and allows US and non-US oil companies with US subsidiaries to begin to explore and advance investment in Venezuela. The waiver is valid for six month until April 2024 and at least one 6-month extension period beyond April 2024 is likely as long as the government of President Maduro adheres to the electoral roadmap agreed with the opposition which establishes a series of electoral guarantees for the presidential election to be held at some point in 2H24. Late 2024 presidential elections in both countries would determine the longer-term fate of US sanctions and Venezuelan oil production.

In terms of oil supply, decades of underinvestment in oil-related infrastructure will likely keep PDVSA's oil production flat throughout the next two years and only short-cycle, capital-light production restoration at Chevron's operations will drive modest growth in Venezuela's oil output. Chevron's oil output rose from 60 kbd earlier this year to around 130 kbd today and will likely get to 150 kbd by year-end. The company plans to add another 65 kbd of oil output by the end of 2024 through its first major drilling campaign in the country (**Figure 21**).

Figure 21: Venezuela crude oil production forecast

kbd



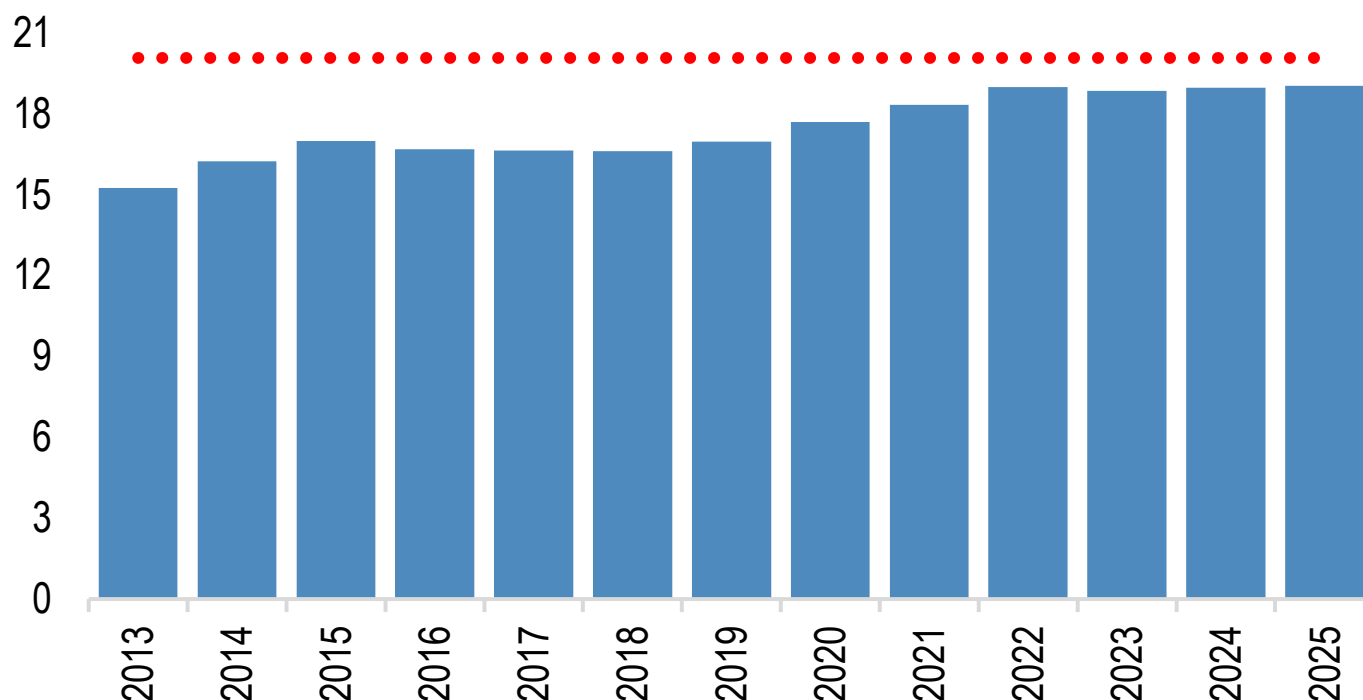
Source: Reuters, J.P. Morgan Commodities research

Refining—demand gains to outstrip supply additions through 2024

Global refining capacity is set to increase by 1.9 mbd in 2024. China will account for only 0.7 mbd of that, as three projects—400 kbd Yulong refinery, 220 kbd Zhenhai refinery and a 120 kbd Ningbo Daxie refinery—commission in 2024. These new additions will bring total Chinese refining capacity to 19.0 mbd, just 1.1 mbd shy of the 20.1 mbd (1 billion ton per annum) planned cap for the nation's refining capacity in 2025 (**Figure 22**). Moreover, in order to reduce carbon emissions, the country has set a minimum size for new oil refineries at 200 kbd. We believe this new rule would likely incentivize Chinese independent refiners to set up facilities in other countries. For example, Sinopec has already set up an entity to invest, build and operate refineries abroad. It broadly mimics China's cap on aluminum smelting capacity of 45 mmta introduced in 2017 post which joint ventures in aluminum smelting were announced between China and Indonesia with the latter becoming a de-facto Chinese growth source while keeping China's carbon balance sheet in check.

Figure 22: China refining capacity

mbd; red line marks the 20.1 mbd refining capacity cap



Source: Wood Mackenzie, S&P Global

Outside of China, Nigeria's 650 kbd Dangote refinery and Mexico's 340 kbd Olmeca refinery will together amount to a 1.0 mbd increase while smaller additions will come from the Middle East and India (**Figure 23**).

While we have 300 kbd of extra refining capacity coming online relative to our total 2024 oil demand growth, technical issues and delays could significantly diminish available capacity. Prime example of technical or operational snags delaying projects are the cases of Nigeria's Dangote refinery and Kuwait's Al Zour refinery. Inaugurated in summer of 2023, Dangote has been unable to secure enough crude cargoes to begin trial runs and is now expected to begin partial operations only in late 2024. Production from Al Zour, too, was delayed by a year from 2021 to 2022. After two years of ramp-up, Al Zour is still facing technical issues with its October production estimated at 325 kbd compared to its 615 kbd total crude processing capacity. Similarly, 340 kbd Olmeca refinery in Mexico is currently operating only at 5% of capacity, well below expectations. The Mexican government is targeting crude processing at Olmeca to rise by about 50% in 2024 and 2025, adding almost 170 kbd of oil products supply.

Our fundamental balances suggest continued tightness in oil products through end-2024, supporting above-average gasoline and diesel cracks. Contingent on a smooth ramp-up of Al Zour in Kuwait and Duqm in Oman as well as smooth commissioning of Dangote in Nigeria and Olmeca in Mexico, balances should loosen in 2025, as refining capacity additions close the gap with demand. As a result, inventories should build back toward more typical levels and cracks should normalize toward historical averages.

Figure 23: Major new refining capacity by region

Refinery	Operator	Location	Capacity (kbd)	Start Date	Status
Duqm	DRPIC	Oman	230	4Q23	Commercial operations have already begun
Dangote	Dangote	Nigeria	650	4Q24	Start up is delayed and is expected to begin work in 2024
Sitra	The Bahrain Petroleum Company BSC	Bahrain	100	1Q24	Refinery capacity to be expanded 100 kbd with start up expected in 2024
Al Zour (P-III)	KIPIC	Kuwait	205	Running	Technical issues keep the refinery from operating at 100%
Karbala	Midland Refineries Co.	Iraq	140	Running	Expected to reach steady state in early 2024
Vizag	HPCL	India	80	1Q24 - 2Q24	Refinery expected to run at full expanded capacity sometime in early 2024
Olmecca	Pemex	Mexico	340	3Q23	Pemex estimates refinery will process 17 kbd in 2023. Ramp-up to continue throughout 2024-2025
Total			1745		

Source: S&P Global

In terms of products, new refining capacity in the Middle East in 2024 will produce additional 215 kbd of diesel, 220 kbd of low-sulfur fuel oil and 145 kbd of jet fuel, gasoline and other products. Most of these volumes are set to be exported to Europe, likely alleviating product shortages in the continent. Stocks of distillates and gasoline in Europe's ARA hub sit currently at the lower end of their five year range. While increased oil products flow into Europe will eventually weigh on European gasoline and distillate cracks, they should remain above their historical averages in 1H24, supported by seasonal increase in demand, still limited increase in supply in 1Q23 and depleted levels of distillates inventories. Heading into the second half of 2024, distillates and gasoline crack may come under pressure moving lower as regional supply-demand imbalance eases.

We estimate that gasoline cracks in Europe are likely to recover from the current \$7/bbl to \$12/bbl in 1Q24, and average \$18-20/bbl over the middle quarters of 2024, before easing to \$5/bbl in 4Q24, closely following seasonal demand patterns (**Figure 24**). We believe European gasoline cracks will likely return to their historical averages in 2025, with risks skewed to lower prices, as the regional demand begins to contract as the electrification of the fleet takes hold and gasoline supply increases. A well-functioning Dangote refinery in Nigeria has the potential to crowd out almost 200-300 kbd of European gasoline exports, leading to an oversupplied European market. Currently, West Africa is home to 25% or 350 kbd of European gasoline exports. We project diesel cracks in Europe to remain largely stable in 2024, easing from the current \$26/bbl to \$22/bbl in 1Q24, \$21/bbl in 2Q before moving higher to \$23/bbl in 3Q and 4Q (**Figure 25**).

Figure 24: Europe gasoline crack, historical and forecast



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

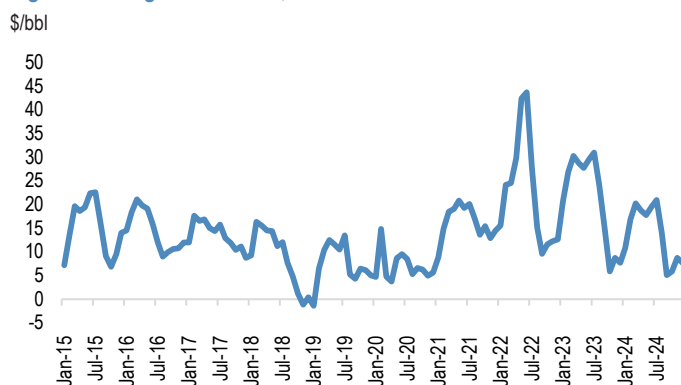
Figure 25: Europe diesel crack, historical and forecast



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

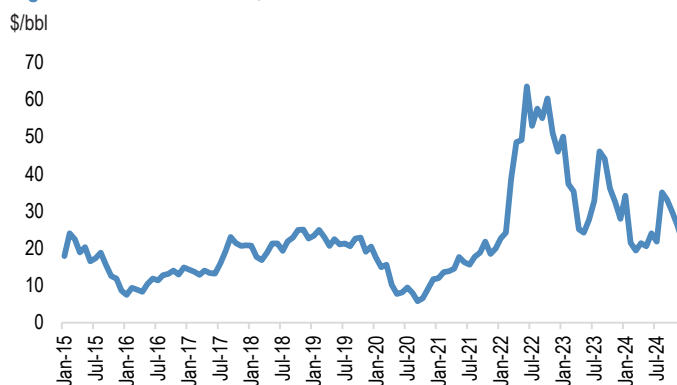
In the US, given our view that US gasoline inventories are likely to remain below their seasonal averages, gasoline cracks are likely to rise from today's \$9/bbl to \$16 in 1Q23, \$19 in 2Q24, ease to \$13 in 3Q24 and return to historical levels of \$7 by 4Q24 (**Figure 26**). US diesel crack, however, should remain elevated throughout 2024 in line with resilient US demand and lower inventories. We estimate US diesel crack will average \$26/bbl in 2024, up \$8/bbl from historical average, with 1Q24 at \$25/bbl, easing to \$22/bbl in 2Q and then averaging \$28/bbl in second half of the year (**Figure 27**).

Figure 26: US gasoline crack, historical and forecast



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 27: US diesel crack, historical and forecast



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

Table 1: Global oil supply and demand balance, 2022E

mbd	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Total Oil Demand	97.8	100.8	99.0	97.0	97.8	100.1	99.4	100.7	100.1	99.5	100.5	100.9	99.1	98.3	100.1	100.3	99.4
Total Oil Supply	97.6	98.1	99.1	98.0	98.4	98.8	100.2	100.0	100.8	101.5	101.0	99.6	98.3	98.4	100.3	100.7	99.4
OPEC Crude	28.3	28.8	28.9	29.0	28.9	29.1	29.4	29.5	29.8	29.9	29.1	29.2	28.7	29.0	29.6	29.4	29.2
OPEC Other Liquids	5.5	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.4	5.3	5.3	5.3	5.3
Non-OPEC Crude + Other	61.3	61.5	62.4	61.1	61.5	61.7	62.8	62.6	63.0	63.7	64.0	62.5	61.7	61.5	62.8	63.4	62.4
Processing Gain	2.6	2.5	2.5	2.6	2.6	2.6	2.6	2.6	2.7	2.6	2.6	2.6	2.5	2.6	2.6	2.6	2.6
Stock Change To Balance	-0.2	-2.7	+0.1	+1.1	+0.6	-1.3	+0.7	-0.7	+0.7	+2.0	+0.5	-1.3	-0.9	+0.1	+0.3	+0.4	-0.0
Stock Change To Balance, mb	-5	-77	+4	+32	+20	-39	+23	-21	+22	+62	+14	-40	-78	+12	+24	+36	-6
Global SPR release, mbd	+0.2	+0.3	+0.4	+1.0	+1.2	+1.4	+1.2	+1.2	+1.4	+1.0	+0.3	+0.5	+0.3	+1.2	+1.3	+0.6	+0.9
Stock Change (Adjusted for SPR, Iran and Russia), mbd	+0.3	-2.8	+1.0	+1.8	+2.2	-0.1	+2.1	+0.5	+2.3	+3.1	+0.9	-0.4	-0.4	+1.3	+1.6	+1.2	+0.9

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 2: Global oil supply and demand balance, 2023E

mbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Total Oil Demand	97.6	101.1	100.3	99.5	101.2	102.2	102.1	103.4	101.7	102.0	102.2	102.7	99.6	101.0	102.4	102.3	101.3
Total Oil Supply	101.0	101.4	101.4	100.6	100.6	100.6	100.2	100.4	102.1	102.2	102.1	101.9	101.3	100.6	100.9	102.1	101.2
OPEC Crude	29.3	29.3	29.3	28.8	28.6	28.6	27.7	28.0	28.2	28.2	28.2	28.2	29.3	28.6	27.9	28.2	28.5
OPEC Other Liquids	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.3	5.3	5.4	5.4	5.4	5.3	5.4
Non-OPEC Crude + Other	63.6	64.0	64.2	63.9	64.1	63.9	64.4	64.3	65.8	65.9	65.9	65.6	64.0	64.0	64.8	65.8	64.6
Processing Gain	2.6	2.6	2.5	2.6	2.5	2.7	2.7	2.8	2.7	2.7	2.7	2.8	2.6	2.6	2.7	2.7	2.7
Stock Change To Balance	+3.4	+0.3	+1.1	+1.1	-0.7	-1.6	-1.9	-3.0	+0.4	+0.1	-0.1	-0.8	+1.6	-0.4	-1.5	-0.3	-0.1
Stock Change To Balance, mb	+105	+8	+33	+34	-20	-47	-58	-93	+12	+4	-2	-25	+5	-34	-139	-24	-192
Global SPR release, mbd	+0.0	+0.0	+0.0	+0.2	+0.3	+0.2	-0.0	-0.1	-0.0	+0.0	-0.1	-0.1	+0.0	+0.3	-0.0	-0.0	+0.0
Stock Change (Adjusted for SPR, Iran and Russia), mbd	+3.3	-0.3	+1.9	+1.2	-0.3	-1.8	-1.9	-3.1	+0.4	+0.1	-0.1	-0.9	+1.7	-0.3	-1.6	-0.3	-0.1

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 3: Global oil supply and demand balance, 2024F

mbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Total Oil Demand	100.7	102.6	101.3	102.1	102.1	102.9	104.4	105.1	102.4	103.4	103.9	104.2	101.5	102.4	104.0	103.8	102.9
Total Oil Supply	102.0	101.9	102.0	102.4	102.8	103.4	103.7	103.5	103.4	103.8	103.9	103.6	102.0	102.9	103.5	103.8	103.0
OPEC Crude	28.3	28.3	28.3	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.3	28.7	28.7	28.7	28.6
OPEC Other Liquids	5.5	5.5	5.4	5.4	5.5	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.5	5.4	5.4	5.4	5.4
Non-OPEC Crude + Other	65.5	65.5	65.6	65.7	66.1	66.6	66.8	66.6	66.5	67.0	67.0	66.7	65.5	66.1	66.7	66.9	66.3
Processing Gain	2.7	2.6	2.6	2.6	2.6	2.7	2.7	2.7	2.7	2.6	2.7	2.7	2.6	2.6	2.7	2.7	2.7
Stock Change To Balance	+1.3	-0.7	+0.7	+0.3	+0.7	+0.5	-0.7	-1.6	+1.0	+0.5	+0.0	-0.6	+0.4	+0.5	-0.4	-0.0	+0.1
Stock Change To Balance, mb	+40	-20	+21	+9	+22	+14	-21	-48	+30	+14	+1	-19	+41	+45	-40	-5	+42
Global SPR release, mbd	-0.1	-0.2	-0.1	-0.1	-0.1	-0.3	-0.2	-0.2	-0.2	+0.0	+0.0	+0.0	-0.2	-0.1	-0.2	+0.0	-0.1
Stock Change (Adjusted for SPR, Iran and Russia), mbd	+1.1	-0.9	+0.5	+0.3	+0.7	+0.2	-0.9	-1.7	+0.8	+0.5	+0.0	-0.6	+0.3	+0.4	-0.6	-0.0	+0.0

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 4: Global oil supply and demand balance, 2025F

mbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Total Oil Demand	102.2	103.6	101.3	103.4	103.1	104.0	106.0	106.1	103.2	104.3	104.6	104.7	102.3	103.5	105.1	104.5	103.9
Total Oil Supply	104.0	104.1	104.2	104.4	104.8	105.1	105.6	105.5	105.2	105.7	105.7	105.4	104.1	104.7	105.4	105.6	105.0
OPEC Crude	28.4	28.4	28.4	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.4	28.3	28.3	28.3	28.3
OPEC Other Liquids	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6
Non-OPEC Crude + Other	67.1	67.3	67.4	67.6	67.9	68.3	68.7	68.6	68.3	68.8	68.8	68.5	67.3	67.9	68.5	68.7	68.1
Processing Gain	2.9	2.9	2.9	2.9	2.9	2.9	2.9	3.0	2.9	2.9	3.0	3.0	2.9	2.9	2.9	2.9	2.9
Stock Change To Balance	+1.8	+0.5	+2.9	+1.0	+1.7	+1.1	-0.4	-0.6	+1.9	+1.4	+1.1	+0.7	+1.8	+1.2	+0.3	+1.1	+1.1
Stock Change To Balance, mb	+55	+15	+90	+29	+52	+32	-13	-20	+58	+44	+32	+23	+160	+112	+25	+99	+396
Global SPR release, mbd	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
Stock Change (Adjusted for SPR, Iran and Russia), mbd	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.2	+0.3	+1.1	+1.1

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 5: J.P. Morgan crude oil price forecasts (US\$/bbl)

Price and forecasts are quarterly and annual spot averages

		2022	1Q23	2Q23	3Q23	4Q23	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025
Brent	Avg	101	82	77	80	85	81	79	84	84	85	83	82	77	73	69	75
	EoP	88	79	72	86	86	86	80	81	88	85	85	79	73	74	67	67
WTI	Avg	97	78	73	76	81	76	75	80	80	81	79	78	73	69	65	71
	EoP	84	78	67	82	81	81	76	77	84	81	81	75	69	70	63	63
WTI - Brent Spread	Avg	-4.4	-4.2	-4.5	-4.0	-4.5	-4.8	-4.5	-4.5	-4.0	-4.5	-4.4	-4.3	-4.3	-4.0	-4.3	-4.2
Realized Avg/ Forward Curve	Brent	99	82	78	86	86	83	81	80	80	79	80	78	77	76	75	76
	WTI	94	76	74	82	83	79	77	76	75	74	76	73	72	71	70	72
Realized EoP/ Forward Curve	Brent	81	79	75	93	78	78	78	77	77	76	76	75	74	73	73	73
	WTI	77	73	70	89	73	73	73	73	72	71	71	70	70	69	68	68

Source: ICE, NYMEX, Bloomberg Finance L.P., J.P. Morgan Commodities Research. Actuals till 2Q23. Forward curve for Brent from 3Q23 and WTI from 2Q23 onwards. Price forecasts last updated on November 17, 2023

Table 6: Global crude oil and products balance, 2022E

mbd	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Crude and Condensate Production	80.6	81.3	81.8	80.5	80.4	80.5	81.7	81.8	82.6	83.3	82.9	82.5	81.2	80.5	82.0	82.9	81.4
Refinery Runs	80.5	80.8	79.6	78.1	79.1	80.8	80.3	80.9	80.3	78.8	81.0	80.9	80.3	79.3	80.5	80.2	79.9
Direct Crude Burn	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0.4	0.4	0.4	0.4	0.6	0.4	0.5
Other Direct Crude Demand	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Crude Oil Demand	81.3	81.6	80.3	78.9	79.9	81.7	81.2	81.9	81.3	79.8	81.8	81.7	81.0	80.2	81.5	81.1	80.7
Crude Balance Change	-0.7	-0.3	1.4	1.6	0.4	-1.2	0.4	0.0	1.3	3.5	1.1	0.8	0.2	0.3	0.6	1.8	0.7
Crude Balance Change, mb*	-21	-8	43	49	13	-35	14	-2	40	108	32	26	15	27	52	166	259
Refinery Processing Gains	2.6	2.5	2.5	2.6	2.6	2.6	2.6	2.6	2.7	2.6	2.6	2.6	2.5	2.6	2.6	2.6	2.6
Refined Products Supply	83.1	83.2	82.1	80.6	81.7	83.4	82.9	83.5	83.0	81.4	83.7	83.5	82.8	81.9	83.1	82.8	82.5
Global NGLs and Non-conventional Oils	17.0	16.8	17.4	17.5	18.0	18.3	18.5	18.2	18.2	18.2	18.1	17.1	17.1	17.9	18.3	17.8	17.7
Global Products & Liquids Supply	97.6	97.6	97.0	95.6	97.2	99.1	98.8	99.1	98.5	97.1	99.1	98.0	97.4	97.3	98.8	98.0	97.6
Global Oil Demand ex Direct Crude Use	97.0	100.0	98.3	96.2	96.9	99.2	98.5	99.7	99.1	98.5	99.7	100.1	98.4	97.4	99.1	99.4	98.3
Product Balance Change	0.5	-2.4	-1.3	-0.6	0.2	-0.1	0.3	-0.6	-0.6	-1.5	-0.6	-2.1	-1.0	-0.2	-0.3	-1.4	-0.7
Product Balance Change, mb*	16	-68	-39	-17	7	-4	10	-19	-17	-45	-18	-65	-91	-14	-27	-128	-260
Total Balance Change	-0.2	-2.7	0.1	1.1	0.6	-1.3	0.7	-0.7	0.7	2.0	0.5	-1.3	-0.9	0.1	0.3	0.4	0.0
Total Balance Change, mb*	-5	-76	4	32	20	-39	23	-21	22	63	15	-40	-77	13	25	38	-1

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 7: Global crude oil and products balance, 2023F

mbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Crude and Condensate Production	83.5	83.9	83.6	82.4	82.1	81.8	81.2	81.5	83.0	83.2	83.3	83.4	83.7	82.1	81.9	83.3	82.5
Refinery Runs	81.1	81.0	81.2	81.4	80.8	81.1	82.6	83.6	81.9	79.8	81.3	84.0	81.1	81.1	82.7	81.7	81.4
Direct Crude Burn	0.4	0.4	0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.6	0.4	0.4	0.4	0.5	0.6	0.5	0.5
Other Direct Crude Demand	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Crude Oil Demand	81.8	81.9	81.9	82.3	81.7	82.1	83.6	84.6	83.0	80.8	82.1	84.8	81.9	82.0	83.7	82.6	82.3
Crude Balance Change	1.7	2.1	1.7	0.1	0.4	-0.3	-2.4	-3.1	0.0	2.4	1.2	-1.4	1.8	0.1	-1.9	0.7	0.2
Crude Balance Change, mb*	52	58	52	3	13	-9	-74	-98	0	74	36	-43	161	8	-171	67	64
Refinery Processing Gains	2.6	2.6	2.5	2.6	2.5	2.7	2.7	2.8	2.7	2.7	2.7	2.8	2.6	2.6	2.7	2.7	2.7
Refined Products Supply	83.7	83.6	83.7	84.1	83.4	83.8	85.4	86.4	84.7	82.5	84.0	86.7	83.7	83.7	85.5	84.4	84.1
Global NGLs and Non-conventional Oils	17.5	17.5	17.8	18.2	18.5	18.9	19.0	19.1	19.1	19.0	18.8	18.5	17.6	18.5	19.1	18.8	18.5
Global Products & Liquids Supply	98.6	98.5	99.0	99.7	99.3	100.0	101.7	102.7	101.0	98.8	100.1	102.5	98.7	99.7	101.8	100.5	99.9
Global Oil Demand ex Direct Crude Use	96.8	100.3	99.5	98.6	100.4	101.3	101.1	102.4	100.6	101.0	101.3	101.9	98.8	100.1	101.4	101.4	100.2
Product Balance Change	1.7	-1.7	-0.6	1.0	-1.1	-1.3	0.5	0.2	0.4	-2.2	-1.3	0.6	-0.1	-0.4	0.4	-1.0	-0.3
Product Balance Change, mb*	54	-49	-18	31	-33	-38	17	8	12	-69	-38	18	-13	-40	37	-88	-105
Total Balance Change	3.4	0.3	1.1	1.1	-0.6	-1.6	-1.8	-2.9	0.4	0.1	-0.1	-0.8	1.6	-0.4	-1.5	-0.2	-0.1
Total Balance Change, mb*	106	9	34	34	-20	-47	-57	-90	12	4	-2	-25	148	-32	-135	-22	-40

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 8: Global crude oil and products balance, 2024F

mbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Crude and Condensate Production	83.6	83.6	83.7	83.8	83.9	84.2	84.4	84.1	84.1	84.7	84.8	84.8	84.6	84.0	84.2	84.8	84.1
Refinery Runs	81.6	82.1	81.9	82.1	82.0	83.4	85.0	85.7	84.0	83.1	85.2	85.8	82.8	82.5	84.9	84.7	83.5
Direct Crude Burn	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Other Direct Crude Demand	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Crude Oil Demand	82.4	82.9	82.7	82.9	82.8	84.2	85.8	86.5	84.8	83.9	86.0	86.6	83.6	83.3	85.7	85.5	84.3
Crude Balance Change	1.2	0.7	0.9	0.9	1.1	0.0	-1.4	-2.3	-0.6	0.7	-1.1	-1.8	1.0	0.7	-1.5	-0.7	-0.2
Crude Balance Change, mb*	36	21	29	28	34	0	-45	-73	-19	22	-34	-57	86	62	-136	-68	-56
Refinery Processing Gains	2.7	2.6	2.6	2.6	2.6	2.7	2.7	2.7	2.7	2.6	2.7	2.7	2.7	2.6	2.7	2.7	2.7
Refined Products Supply	84.3	84.7	84.5	84.7	84.6	86.1	87.7	88.4	86.6	85.8	87.9	88.6	85.5	85.1	87.6	87.4	86.2
Global NGLs and Non-conventional Oils	18.4	18.3	18.3	18.6	18.9	19.2	19.4	19.4	19.2	19.2	19.1	18.8	18.6	18.9	19.3	19.0	18.9
Global Products & Liquids Supply	100.0	100.4	100.2	100.7	101.0	102.6	104.4	105.1	103.2	102.3	104.2	104.6	101.3	101.4	104.2	103.7	102.4
Global Oil Demand ex Direct Crude Use	99.9	101.8	100.5	101.3	101.3	102.1	103.6	104.3	101.6	102.6	103.1	103.4	101.8	101.6	103.2	103.0	102.1
Product Balance Change	0.2	-1.4	-0.2	-0.6	-0.4	0.5	0.8	0.8	1.6	-0.2	1.2	1.2	-0.5	-0.2	1.1	0.7	0.3
Product Balance Change, mb*	5	-40	-7	-17	-11	14	24	25	49	-7	35	38	-43	-14	99	66	107
Total Balance Change	1.3	-0.7	0.7	0.3	0.7	0.5	-0.7	-1.5	1.0	0.5	0.0	-0.6	0.5	0.5	-0.4	0.0	0.1
Total Balance Change, mb*	41	-19	22	10	23	15	-20	-48	30	15	1	-18	43	48	-37	-2	51

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 9: Global crude oil and products balance, 2025F

mbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Crude and Condensate Production	85.0	85.1	85.2	85.0	85.0	85.1	85.4	85.2	85.1	85.7	85.8	85.8	83.2	85.0	86.2	85.8	85.0
Refinery Runs	84.9	84.1	83.8	83.3	83.3	84.7	85.6	86.3	84.8	83.8	85.8	86.6	82.4	83.8	86.5	85.4	84.5
Direct Crude Burn	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Other Direct Crude Demand	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Crude Oil Demand	85.7	84.9	84.6	84.1	84.1	85.5	86.4	87.1	85.6	84.6	86.6	87.4	83.2	84.6	87.3	86.2	85.3
Crude Balance Change	-0.7	0.2	0.6	0.9	0.9	-0.5	-1.1	-1.9	-0.5	1.1	-0.8	-1.6	0.0	0.5	-1.2	-0.4	-0.3
Crude Balance Change, mb*	-22	5	19	27	29	-14	-33	-57	-16	34	-24	-49	2	42	-107	-39	-101
Refinery Processing Gains	2.9	2.9	2.9	2.9	2.9	2.9	2.9	3.0	2.9	2.9	3.0	3.0	2.8	2.9	3.0	2.9	2.9
Refined Products Supply	87.8	87.0	86.6	86.1	86.2	87.6	88.6	89.2	87.8	86.7	88.7	89.6	85.2	86.6	89.5	88.3	87.4
Global NGLs and Non-conventional Oils	19.1	19.0	19.1	19.4	19.8	20.0	20.3	20.3	20.1	20.1	19.9	19.6	18.6	19.7	20.4	19.9	19.7
Global Products & Liquids Supply	104.0	103.1	102.8	102.7	103.1	104.8	105.9	106.6	104.9	103.8	105.7	106.2	101.1	103.5	107.0	105.2	104.2
Global Oil Demand ex Direct Crude Use	101.4	102.7	100.5	102.6	102.3	103.2	105.2	105.3	102.4	103.5	103.8	103.9	99.3	102.7	105.5	103.7	102.8
Product Balance Change	2.5	0.4	2.3	0.1	0.8	1.5	0.7	1.2	2.5	0.4	1.9	2.3	1.7	0.8	1.5	1.5	1.4
Product Balance Change, mb*	78	10	72	3	24	46	21	39	76	11	57	73	160	73	135	141	508
Total Balance Change	1.8	0.6	2.9	1.0	1.7	1.1	-0.4	-0.6	2.0	1.4	1.1	0.8	1.8	1.3	0.3	1.1	1.1
Total Balance Change, mb*	56	16	91	30	53	33	-12	-19	59	45	33	24	162	115	28	102	407

Source: IEA, EIA, Wood Mackenzie, Rystad, Genscape, FlightAware, MariTrace, Google, TomTom, US Department of Transportation, J.P. Morgan Commodities Research

Table 10: OPEC crude oil production, 2022E

kdb	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Algeria	970	970	970	990	1,000	1,010	1,020	1,020	1,030	1,030	1,040	1,020	960	977	1,010	1,027	912
Angola	1,150	1,120	1,180	1,160	1,160	1,160	1,220	1,170	1,190	1,150	1,090	1,090	1,117	1,152	1,180	1,170	1,126
Congo	270	260	270	260	270	280	290	270	260	270	270	260	273	263	280	267	267
Eq. Guinea	90	100	90	90	90	100	100	110	100	100	70	60	90	93	97	103	102
Gabon	200	190	190	190	180	180	190	220	200	210	220	210	190	190	183	210	184
Iran	2,500	2,520	2,560	2,550	2,580	2,580	2,580	2,580	2,580	2,550	2,560	2,560	2,507	2,543	2,580	2,570	2,415
Iraq	4,310	4,260	4,260	4,340	4,420	4,390	4,380	4,450	4,420	4,500	4,580	4,490	4,243	4,288	4,397	4,456	4,023
Kuwait	2,550	2,570	2,600	2,620	2,640	2,690	2,720	2,770	2,810	2,820	2,800	2,680	2,530	2,597	2,683	2,800	2,415
Libya	1,050	990	1,120	1,070	900	770	650	650	1,100	1,070	1,180	1,140	1,106	1,058	773	939	1,139
Nigeria	1,380	1,570	1,550	1,520	1,400	1,230	1,300	1,230	1,100	1,070	1,110	1,170	1,396	1,547	1,309	1,134	1,458
Neutral Zone	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Saudi Arabia	9,950	10,080	10,250	10,250	10,400	10,450	10,550	10,770	10,920	11,020	10,900	10,460	9,877	10,191	10,466	10,902	9,117
UAE	3,000	3,025	3,075	3,125	3,250	3,300	3,325	3,450	3,150	3,350	3,400	3,300	2,950	3,075	3,292	3,316	2,774
Venezuela	750	630	680	720	750	750	750	700	670	670	700	660	673	677	750	680	560
Azerbaijan	597	590	600	580	580	570	580	550	550	540	540	550	591	590	577	547	594
Bahrain	180	190	180	200	200	170	180	200	200	200	210	200	87	190	183	200	50
Brunei	90	90	90	80	80	80	70	70	70	60	60	50	90	87	77	67	87
Kazakhstan	1,680	1,600	1,650	1,550	1,330	1,520	1,220	1,390	1,240	1,220	1,400	1,580	1,630	1,598	1,358	1,284	1,490
Malaysia	420	420	420	430	400	400	400	400	350	360	400	390	407	423	400	370	427
Mexico	1,671	1,681	1,662	1,655	1,636	1,644	1,652	1,651	1,654	1,649	1,651	1,634	1,673	1,666	1,644	1,651	1,691
Oman	800	810	820	830	840	840	850	870	880	880	880	850	783	820	843	877	755
Russia	10,010	10,080	10,110	10,040	9,140	9,290	9,750	9,800	9,770	9,800	9,850	9,870	9,990	10,076	9,392	9,790	9,656
Sudan	60	60	60	60	60	60	60	60	60	60	60	60	57	60	60	60	62
S. Sudan	140	150	140	140	150	150	150	160	160	150	150	150	133	143	150	157	146
OPEC 10	23,870	24,145	24,435	24,545	24,810	24,790	25,095	25,460	25,180	25,520	25,480	24,740	23,626	24,373	24,897	25,385	22,378
OPEC 13	28,170	28,285	28,795	28,885	29,040	28,890	29,075	29,390	29,530	29,810	29,920	29,100	27,913	28,650	29,000	29,574	26,493
Non-OPEC	15,648	15,671	15,732	15,565	14,416	14,724	14,912	15,151	14,934	14,919	15,201	15,334	15,440	15,654	14,684	15,002	14,959
OPEC+	39,518	39,816	40,167	40,110	39,226	39,514	40,007	40,611	40,114	40,439	40,681	40,074	39,066	40,027	39,582	40,387	37,337

Source: Rystad, Platts, Refinitiv, J.P. Morgan Commodities Research

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Global Commodities Research
 Oil Outlook 2024/2025
 20 November 2023

J.P.Morgan

Table 11: OPEC crude oil production, 2023E

kbbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Algeria	1,010	1,010	1,010	1,000	1,000	960	940	940	930	950	950	950	1,023	1,007	967	940	1,009
Angola	1,120	1,120	1,050	1,010	1,090	1,120	1,110	1,120	1,110	1,130	1,150	1,100	1,100	1,060	1,107	1,120	1,151
Congo	220	260	270	280	280	290	290	290	270	250	260	262	250	270	287	270	265
Eq. Guinea	60	50	50	50	60	60	60	60	60	50	50	50	63	50	60	57	89
Gabon	180	180	190	200	200	200	200	210	210	210	210	204	203	190	200	210	197
Iran	2,580	2,610	2,620	2,600	2,620	2,660	2,710	2,760	2,950	3,010	3,050	3,050	2,567	2,610	2,663	2,906	2,565
Iraq	4,460	4,480	4,330	4,390	4,100	4,100	4,130	4,210	4,320	4,300	4,400	4,400	4,510	4,402	4,110	4,276	4,413
Kuwait	2,670	2,680	2,680	2,680	2,680	2,550	2,550	2,550	2,540	2,550	2,550	2,550	2,717	2,680	2,593	2,547	2,700
Libya	1,170	1,160	1,170	1,150	1,140	1,150	1,140	1,120	1,120	1,130	1,120	1,100	1,164	1,160	1,143	1,123	984
Nigeria	1,330	1,400	1,450	1,430	1,180	1,400	1,420	1,320	1,380	1,480	1,460	1,450	1,204	1,426	1,334	1,392	1,297
Neutral Zone	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Saudi Arabia	10,480	10,420	10,460	10,450	10,500	10,000	9,990	9,005	8,950	9,010	9,000	9,000	10,615	10,443	10,162	8,988	10,546
UAE	3,250	3,250	3,375	3,350	3,200	3,300	3,275	3,275	3,350	3,325	3,225	3,250	3,317	3,323	3,259	3,317	3,251
Venezuela	660	670	690	690	710	770	800	810	790	770	760	790	673	683	760	790	695
Azerbaijan	550	530	531	520	510	500	500	500	500	491	480	500	547	527	503	497	565
Bahrain	200	150	190	190	190	190	200	200	200	200	200	200	203	176	193	200	194
Brunei	70	80	80	70	70	70	70	70	70	60	60	70	60	77	70	67	72
Kazakhstan	1,600	1,600	1,550	1,600	1,600	1,550	1,550	1,500	1,450	1,540	1,560	1,600	1,526	1,584	1,566	1,496	1,441
Malaysia	400	400	400	390	390	380	370	370	370	370	370	370	397	397	380	370	397
Mexico	1,645	1,674	1,669	1,680	1,693	1,701	1,698	1,661	1,687	1,696	1,687	1,659	1,644	1,675	1,697	1,681	1,651
Oman	840	840	840	840	840	840	800	800	800	800	800	810	857	840	827	800	849
Russia	9,860	9,850	9,949	9,600	9,650	9,600	9,600	9,550	9,500	9,500	9,500	9,500	9,860	9,795	9,616	9,517	9,779
Sudan	60	60	50	50	50	50	50	50	50	50	50	50	60	53	50	50	60
S. Sudan	150	150	140	140	140	140	150	150	150	150	160	150	150	143	143	150	150
OPEC 10	24,780	24,850	24,865	24,840	24,290	23,980	23,965	22,980	23,120	23,255	23,255	23,216	25,003	24,851	24,077	23,117	24,918
OPEC 13	29,190	29,290	29,345	29,280	28,760	28,560	28,615	27,670	27,980	28,165	28,185	28,156	29,407	29,304	28,644	27,936	29,161
Non-OPEC	15,375	15,334	15,399	15,080	15,133	15,021	14,988	14,851	14,777	14,857	14,867	14,909	15,303	15,267	15,047	14,828	15,159
OPEC+	40,155	40,184	40,264	39,920	39,423	39,001	38,953	37,831	37,897	38,112	38,122	38,125	40,306	40,118	39,124	37,945	40,077

Source: Rystad, Platts, Refinitiv, J.P. Morgan Commodities Research

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Global Commodities Research

J.P.Morgan

20 November 2023

Table 12: OPEC crude oil production, 2024F

kbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Algeria	950	950	950	950	950	950	950	950	950	950	950	950	950	950	950	950	966
Angola	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,117	1,100	1,100	1,100	1,101
Congo	259	250	250	250	250	240	240	240	240	240	240	240	260	250	243	240	272
Eq. Guinea	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	54
Gabon	202	199	196	197	196	195	193	193	192	191	190	189	205	197	195	192	201
Iran	3,100	3,100	3,100	3,100	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,067	3,100	3,150	3,150	2,813
Iraq	4,400	4,400	4,400	4,400	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,400	4,400	4,450	4,450	4,297
Kuwait	2,550	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,550	2,560	2,560	2,560	2,592
Libya	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,107	1,100	1,100	1,100	1,133
Nigeria	1,450	1,400	1,400	1,400	1,400	1,400	1,400	1,400	1,400	1,400	1,400	1,400	1,453	1,400	1,400	1,400	1,401
Neutral Zone	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Saudi Arabia	9,000	9,000	9,000	9,000	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,000	9,000	9,250	9,250	9,642
UAE	3,275	3,350	3,350	3,350	3,350	3,350	3,350	3,400	3,400	3,400	3,400	3,400	3,250	3,350	3,350	3,400	3,287
Venezuela	800	830	850	850	875	875	900	900	900	900	900	900	783	843	883	900	755
Azerbaijan	500	500	500	500	500	500	500	500	500	500	500	500	493	500	500	500	505
Bahrain	200	190	190	190	190	190	190	190	190	190	190	190	200	190	190	190	192
Brunei	70	70	70	70	70	70	70	70	70	70	70	70	67	70	70	70	70
Kazakhstan	1,600	1,620	1,620	1,620	1,620	1,620	1,620	1,500	1,340	1,500	1,630	1,630	1,587	1,620	1,620	1,446	1,558
Malaysia	370	400	400	400	400	400	400	400	400	400	400	400	370	400	400	400	379
Mexico	1,625	1,638	1,608	1,588	1,594	1,597	1,594	1,631	1,624	1,625	1,620	1,612	1,657	1,611	1,595	1,627	1,677
Oman	810	810	810	810	810	810	810	810	810	810	810	810	807	810	810	810	818
Russia	9,500	9,500	9,500	9,500	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,500	9,500	9,650	9,650	9,606
Sudan	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	51
S. Sudan	150	150	150	150	150	150	150	150	150	150	150	150	153	150	150	150	148
OPEC 10	23,235	23,259	23,256	23,257	23,556	23,545	23,543	23,593	23,592	23,591	23,590	23,589	23,236	23,257	23,548	23,592	23,814
OPEC 13	28,235	28,289	28,306	28,307	28,681	28,670	28,693	28,743	28,742	28,741	28,740	28,739	28,193	28,300	28,681	28,742	28,514
Non-OPEC	14,875	14,928	14,898	14,878	15,034	15,037	15,034	14,951	14,784	14,945	15,070	15,062	14,883	14,901	15,035	14,893	15,005
OPEC+	38,110	38,187	38,154	38,135	38,590	38,582	38,578	38,544	38,376	38,536	38,661	38,651	38,119	38,159	38,583	38,485	38,819

Source: Rystad, Platts, Refinitiv, J.P. Morgan Commodities Research

Table 13: OPEC crude oil production, 2025F

kbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Algeria	950	950	950	950	950	950	950	950	950	950	950	950	950	950	950	950	950
Angola	961	958	955	941	934	921	908	895	886	887	876	864	958	932	897	875	915
Congo	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240
Eq. Guinea	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
Gabon	184	181	179	178	176	176	177	175	173	170	168	165	181	177	175	168	175
Iran	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150	3,150
Iraq	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450	4,450
Kuwait	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560
Libya	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100
Nigeria	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300
Neutral Zone	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Saudi Arabia	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250	9,250
UAE	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230	3,230
Venezuela	950	950	950	950	950	950	975	975	975	975	975	975	950	950	975	975	963
Azerbaijan	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500
Bahrain	190	190	190	190	190	190	190	190	190	190	190	190	190	190	190	190	190
Brunei	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70
Kazakhstan	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630	1,630
Malaysia	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
Mexico	1,582	1,554	1,529	1,529	1,525	1,506	1,511	1,499	1,489	1,487	1,476	1,452	1,555	1,520	1,500	1,472	1,512
Oman	810	810	810	810	810	810	810	810	810	810	810	810	810	810	810	810	810
Russia	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650	9,650
Sudan	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
S. Sudan	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
OPEC 10	23,175	23,168	23,164	23,149	23,141	23,127	23,115	23,100	23,089	23,087	23,074	23,059	23,169	23,139	23,102	23,073	23,120
OPEC 13	28,375	28,368	28,364	28,349	28,341	28,327	28,340	28,325	28,314	28,312	28,299	28,284	28,369	28,339	28,327	28,298	28,333
Non-OPEC	15,032	15,004	14,979	14,979	14,975	14,956	14,961	14,949	14,939	14,937	14,926	14,902	15,005	14,970	14,950	14,922	14,962
OPEC+	38,207	38,173	38,143	38,129	38,116	38,083	38,077	38,049	38,028	38,025	38,000	37,961	38,174	38,109	38,052	37,995	38,082

Source: Rystad, Platts, Refinitiv, J.P. Morgan Commodities Research

Table 14: Global crude oil supply, 2022E

mbd	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	45.2	45.4	45.7	44.4	44.3	44.4	45.2	45.3	45.6	46.2	46.4	46.0	45.4	44.4	45.4	46.2	45.4
Africa	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Asia	6.0	6.1	6.0	5.9	6.0	6.0	5.8	5.7	5.8	5.8	5.9	5.8	6.0	6.0	5.8	5.8	5.9
China	4.1	4.1	4.1	4.1	4.1	4.1	4.0	3.9	4.0	4.0	4.0	3.9	4.1	4.1	4.0	4.0	4.0
FSU	12.5	12.6	12.5	11.3	11.7	11.8	12.0	11.8	11.8	12.1	12.3	12.3	12.5	11.6	11.9	12.2	12.1
Russia	10.1	10.1	10.0	9.1	9.3	9.8	9.8	9.8	9.8	9.9	9.9	9.9	10.1	9.4	9.8	9.9	9.8
Latin America	4.8	4.9	4.9	5.0	5.0	4.9	5.1	5.3	5.4	5.5	5.3	5.3	4.9	5.0	5.3	5.4	5.1
Brazil	2.9	2.8	2.9	2.9	2.8	2.7	2.9	3.0	3.0	3.2	3.0	3.0	2.9	2.8	3.0	3.0	2.9
Middle East	1.7	1.7	1.7	1.7	1.7	1.7	1.8	1.8	1.8	1.8	1.7	1.7	1.7	1.7	1.8	1.7	1.7
Europe	2.8	2.9	2.8	2.7	2.6	2.3	2.6	2.7	2.5	2.7	2.8	2.8	2.8	2.6	2.6	2.8	2.7
Norway	1.7	1.8	1.7	1.7	1.6	1.3	1.6	1.8	1.6	1.8	1.8	1.8	1.8	1.5	1.7	1.8	1.7
North America	16.1	16.1	16.6	16.4	16.2	16.4	16.7	16.8	17.1	17.1	17.3	16.9	16.3	16.3	16.9	17.1	16.6
United States	10.3	10.0	10.5	10.5	10.5	10.6	10.6	10.7	11.0	11.0	11.0	10.8	10.3	10.5	10.8	11.0	10.6
Canada	4.2	4.4	4.4	4.3	4.1	4.2	4.4	4.4	4.5	4.4	4.6	4.4	4.3	4.2	4.4	4.5	4.4
OECD Pacific	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
OPEC	28.3	28.8	28.9	29.0	28.9	29.1	29.4	29.5	29.8	29.9	29.1	29.2	28.7	29.0	29.6	29.4	29.2
East of Suez	22.5	22.7	22.9	23.3	23.4	23.6	24.0	23.9	24.2	24.2	23.5	23.4	22.7	23.4	24.0	23.7	23.5
Africa	5.2	5.4	5.3	5.0	4.7	4.8	4.7	5.0	4.9	5.0	5.0	5.1	5.3	4.8	4.8	5.0	5.0
Latin America	0.6	0.7	0.7	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.7	0.7	0.7
Total	73.5	74.2	74.6	73.4	73.2	73.4	74.6	74.9	75.4	76.1	75.5	75.2	74.1	73.4	75.0	75.6	74.5

Source: Rystad, J.P. Morgan Commodities Research

Table 15: Global crude oil supply, 2023E

mbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	46.6	46.6	46.5	46.0	45.9	45.6	45.9	45.9	47.0	47.2	47.5	47.4	46.6	45.8	46.3	47.4	46.5
Africa	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Asia	6.0	6.0	6.0	6.0	6.0	6.0	5.8	5.8	5.9	5.9	5.9	5.8	6.0	6.0	5.8	5.8	5.9
China	4.2	4.2	4.2	4.1	4.2	4.2	4.0	4.0	4.1	4.1	4.1	4.0	4.2	4.2	4.0	4.1	4.1
FSU	12.2	12.3	12.0	12.0	11.9	11.9	11.8	11.7	11.8	11.8	11.8	11.8	12.2	11.9	11.8	11.8	11.9
Russia	9.9	9.9	9.6	9.7	9.6	9.6	9.6	9.5	9.5	9.5	9.5	9.5	9.8	9.6	9.5	9.5	9.6
Latin America	5.5	5.5	5.4	5.4	5.5	5.6	5.8	5.8	6.0	5.9	6.0	6.0	5.5	5.5	5.9	6.0	5.7
Brazil	3.2	3.2	3.0	3.1	3.1	3.3	3.4	3.5	3.6	3.5	3.6	3.6	3.1	3.1	3.5	3.6	3.3
Middle East	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Europe	2.7	2.7	2.8	2.7	2.6	2.7	2.8	2.7	2.8	2.8	2.8	2.8	2.7	2.7	2.7	2.8	2.7
Norway	1.7	1.7	1.8	1.7	1.7	1.8	1.8	1.7	1.8	1.8	1.8	1.8	1.7	1.7	1.7	1.8	1.8
North America	17.3	17.3	17.5	17.1	17.0	16.5	16.8	16.9	17.8	18.0	18.1	18.1	17.4	16.9	17.2	18.1	17.4
United States	11.2	11.1	11.4	11.2	11.3	11.5	11.5	11.6	11.6	11.6	11.7	11.7	11.2	11.3	11.6	11.7	11.4
Canada	4.4	4.5	4.5	4.2	4.1	3.3	3.7	3.7	4.5	4.7	4.8	4.8	4.5	3.9	3.9	4.7	4.3
OECD Pacific	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
OPEC	29.3	29.3	29.3	28.8	28.6	28.6	27.7	28.0	28.2	28.2	28.2	28.2	29.3	28.6	27.9	28.2	28.5
East of Suez	23.4	23.5	23.5	23.1	22.6	22.7	21.8	22.1	22.2	22.2	22.3	22.3	23.5	22.8	22.0	22.3	22.6
Africa	5.2	5.2	5.1	5.0	5.2	5.2	5.1	5.1	5.2	5.2	5.1	5.1	5.2	5.1	5.1	5.1	5.1
Latin America	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.8	0.8	0.8
Total	75.9	76.0	75.8	74.8	74.5	74.2	73.6	73.9	75.2	75.4	75.6	75.7	75.9	74.5	74.2	75.6	75.0

Source: Rystad, J.P. Morgan Commodities Research

Table 16: Global crude oil supply, 2024F

mbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	47.5	47.5	47.5	47.3	47.4	47.7	47.8	47.5	47.6	48.1	48.3	48.3	47.5	47.5	47.6	48.2	47.7
Africa	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Asia	5.9	5.9	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	5.9	6.0	6.0	6.0	6.0
China	4.1	4.1	4.1	4.1	4.1	4.1	4.2	4.2	4.2	4.2	4.2	4.1	4.1	4.1	4.2	4.2	4.1
FSU	11.9	11.9	11.9	12.0	12.0	12.0	11.9	11.7	11.9	12.0	12.0	12.0	11.9	12.0	11.8	12.0	11.9
Russia	9.5	9.5	9.5	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.5	9.7	9.7	9.7	9.6
Latin America	6.0	6.0	6.0	6.0	6.1	6.1	6.2	6.2	6.2	6.1	6.1	6.2	6.0	6.1	6.2	6.1	6.1
Brazil	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Middle East	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Europe	2.8	2.7	2.7	2.7	2.7	2.7	2.7	2.6	2.6	2.7	2.8	2.7	2.7	2.7	2.6	2.7	2.7
Norway	1.8	1.8	1.8	1.8	1.7	1.7	1.7	1.7	1.6	1.7	1.8	1.7	1.8	1.7	1.7	1.7	1.7
North America	18.1	18.1	18.1	17.8	17.8	18.0	18.2	18.2	18.2	18.4	18.5	18.6	18.1	17.9	18.2	18.5	18.2
United States	11.7	11.7	11.7	11.7	11.7	11.7	11.8	11.8	11.9	11.9	12.0	12.0	11.7	11.7	11.8	12.0	11.8
Canada	4.8	4.8	4.8	4.4	4.4	4.6	4.7	4.7	4.6	4.8	4.8	4.9	4.8	4.5	4.7	4.8	4.7
OECD Pacific	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
OPEC	28.3	28.3	28.3	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.7	28.3	28.7	28.7	28.7	28.6
East of Suez	22.4	22.4	22.4	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.4	22.8	22.8	22.8	22.7
Africa	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Latin America	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.9	0.9	0.9	0.9
Total	75.7	75.8	75.8	76.0	76.1	76.4	76.5	76.3	76.4	76.8	77.0	77.0	75.8	76.1	76.4	77.0	76.3

Source: Rystad, J.P. Morgan Commodities Research

Table 17: Global crude oil supply, 2025F

mbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	48.7	48.8	48.9	48.7	48.8	48.8	49.1	49.0	48.9	49.5	49.6	49.7	48.8	48.8	49.0	49.6	49.0
Africa	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Asia	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	5.9	6.0	6.0	6.0	6.0	6.0
China	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.1	4.1	4.1	4.2	4.2	4.2	4.1	4.2
FSU	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Russia	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Latin America	6.2	6.3	6.3	6.4	6.5	6.5	6.6	6.7	6.8	6.8	6.9	7.0	6.3	6.5	6.7	6.9	6.6
Brazil	3.6	3.6	3.7	3.8	3.8	3.9	3.9	4.0	4.1	4.1	4.2	4.2	3.6	3.8	4.0	4.2	3.9
Middle East	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Europe	2.9	2.9	2.9	3.0	2.9	2.8	3.0	2.9	2.7	2.9	3.0	3.0	2.9	2.9	2.9	3.0	2.9
Norway	1.9	1.9	1.9	2.0	1.9	1.9	2.1	2.0	1.8	2.0	2.0	2.1	1.9	1.9	2.0	2.0	2.0
North America	18.7	18.8	18.7	18.4	18.5	18.6	18.5	18.6	18.6	18.8	18.8	18.8	18.7	18.5	18.6	18.8	18.7
United States	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.2	12.2	12.2	12.2	12.2	12.1	12.1	12.2	12.2	12.2
Canada	5.0	5.0	4.9	4.6	4.7	4.8	4.7	4.7	4.6	4.9	4.9	4.9	5.0	4.7	4.7	4.9	4.8
OECD Pacific	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
OPEC	28.4	28.4	28.4	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.3	28.4	28.3	28.3	28.3	28.3
East of Suez	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6
Africa	4.8	4.8	4.8	4.8	4.8	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.8	4.7	4.7	4.7	4.7
Latin America	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Total	77.0	77.2	77.2	77.1	77.1	77.2	77.4	77.3	77.2	77.8	77.9	77.9	77.2	77.1	77.3	77.9	77.4

Source: Rystad, J.P. Morgan Commodities Research

Table 18: Global liquids supply, 2022E

mbd	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	61.3	61.5	62.4	61.1	61.5	61.7	62.8	62.6	63.0	63.7	64.0	62.5	61.7	61.5	62.8	63.4	62.4
Africa	1.3	1.2	1.3	1.3	1.2	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Asia	6.7	6.8	6.7	6.6	6.7	6.7	6.5	6.4	6.5	6.5	6.6	6.5	6.7	6.7	6.5	6.5	6.6
China	4.2	4.2	4.3	4.2	4.2	4.3	4.1	4.1	4.2	4.1	4.2	4.1	4.2	4.2	4.1	4.1	4.2
FSU	14.3	14.4	14.2	13.1	13.3	13.5	13.7	13.4	13.6	13.7	14.1	14.1	14.3	13.3	13.6	14.0	13.8
Russia	11.3	11.4	11.3	10.4	10.5	10.9	11.0	10.9	11.1	11.0	11.2	11.2	11.3	10.6	11.0	11.1	11.0
Latin America	5.2	5.2	5.2	5.4	5.3	5.2	5.4	5.6	5.7	5.8	5.7	5.7	5.2	5.3	5.6	5.7	5.5
Brazil	3.0	2.9	3.0	3.0	2.9	2.8	3.0	3.1	3.1	3.2	3.1	3.1	3.0	2.9	3.1	3.1	3.0
Middle East	3.1	3.1	3.1	3.1	3.1	3.1	3.2	3.2	3.2	3.2	3.2	3.1	3.1	3.1	3.2	3.2	3.1
Europe	3.3	3.3	3.3	3.1	3.0	2.7	3.0	3.1	2.9	3.1	3.2	3.2	3.3	2.9	3.0	3.2	3.1
Norway	2.0	2.0	2.0	1.9	1.8	1.5	1.9	2.0	1.8	2.0	2.0	2.0	2.0	1.7	1.9	2.0	1.9
North America	24.3	24.3	25.4	25.1	24.9	25.2	25.6	25.6	26.0	26.2	26.4	25.4	24.7	25.1	25.8	26.0	25.4
United States	17.0	16.8	17.7	17.7	17.7	17.8	18.0	18.0	18.5	18.5	18.5	17.7	17.2	17.7	18.2	18.3	17.8
Canada	5.3	5.5	5.6	5.4	5.2	5.4	5.6	5.6	5.5	5.7	5.8	5.6	5.5	5.3	5.6	5.7	5.5
OECD Pacific	0.4	0.4	0.4	0.4	0.5	0.4	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Other Liquids	2.8	2.8	2.8	3.0	3.5	3.6	3.7	3.6	3.4	3.4	3.3	2.9	2.8	3.4	3.6	3.2	3.2
OPEC	33.7	34.1	34.2	34.4	34.2	34.4	34.7	34.8	35.1	35.2	34.4	34.5	34.0	34.3	34.9	34.7	34.5
East of Suez	22.5	22.7	22.9	23.3	23.4	23.6	24.0	23.9	24.2	24.2	23.5	23.4	22.7	23.4	24.0	23.7	23.5
Africa	5.2	5.4	5.3	5.0	4.7	4.8	4.7	5.0	4.9	5.0	5.0	5.1	5.3	4.8	4.8	5.0	5.0
Latin America	0.6	0.7	0.7	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.7	0.7	0.7
NGL & other liquids	5.5	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.4	5.3	5.3	5.3	5.3
Processing gains	2.6	2.5	2.5	2.6	2.6	2.6	2.6	2.6	2.7	2.6	2.6	2.6	2.5	2.6	2.6	2.6	2.6
Total	97.6	98.1	99.1	98.0	98.4	98.8	100.2	100.0	100.8	101.5	101.0	99.6	98.3	98.4	100.3	100.7	99.4

Source: Rystad, J.P. Morgan Commodities Research

Table 19: Global liquids supply, 2023F

mbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	63.6	64.0	64.2	63.9	64.1	63.9	64.4	64.3	65.8	65.9	65.9	65.6	64.0	64.0	64.8	65.8	64.6
Africa	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Asia	6.7	6.7	6.8	6.7	6.7	6.7	6.6	6.6	6.6	6.6	6.5	6.5	6.7	6.7	6.6	6.5	6.6
China	4.3	4.3	4.4	4.3	4.3	4.3	4.2	4.2	4.2	4.2	4.2	4.2	4.3	4.3	4.2	4.2	4.3
FSU	14.0	14.3	14.0	13.8	13.8	13.7	13.6	13.5	13.7	13.7	13.7	13.7	14.1	13.8	13.6	13.7	13.8
Russia	11.1	11.5	11.1	11.0	11.0	10.9	10.8	10.8	10.9	10.9	10.9	10.9	11.2	10.9	10.8	10.9	11.0
Latin America	5.9	5.8	5.7	5.8	5.8	6.0	6.1	6.1	6.3	6.2	6.3	6.4	5.8	5.8	6.2	6.3	6.0
Brazil	3.3	3.3	3.1	3.1	3.2	3.4	3.5	3.4	3.7	3.6	3.7	3.7	3.2	3.2	3.5	3.6	3.4
Middle East	3.1	3.2	3.2	3.2	3.1	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Europe	3.1	3.1	3.2	3.1	3.1	3.1	3.2	3.1	3.2	3.3	3.3	3.2	3.2	3.1	3.1	3.2	3.2
Norway	1.9	2.0	2.0	2.0	1.9	2.0	2.0	1.9	2.0	2.0	2.1	2.0	2.0	2.0	2.0	2.0	2.0
North America	26.1	26.2	26.8	26.5	26.3	26.0	26.3	26.5	27.6	27.8	27.9	27.9	26.4	26.3	26.8	27.9	26.8
United States	18.4	18.5	19.0	19.0	19.1	19.4	19.4	19.5	19.8	19.8	19.9	19.9	18.6	19.2	19.6	19.9	19.3
Canada	5.6	5.6	5.7	5.4	5.1	4.4	4.8	4.8	5.6	5.9	6.0	6.0	5.7	5.0	5.1	6.0	5.4
OECD Pacific	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Other Liquids	2.9	2.9	3.0	3.1	3.7	3.7	3.9	3.8	3.7	3.6	3.4	3.1	2.9	3.5	3.8	3.4	3.4
OPEC	34.7	34.8	34.7	34.1	33.9	34.0	33.1	33.4	33.5	33.6	33.4	33.6	34.7	34.0	33.3	33.5	33.9
East of Suez	23.4	23.5	23.5	23.1	22.6	22.7	21.8	22.1	22.2	22.2	22.3	22.3	23.5	22.8	22.0	22.3	22.6
Africa	5.2	5.2	5.1	5.0	5.2	5.2	5.1	5.1	5.2	5.2	5.1	5.1	5.2	5.1	5.1	5.1	5.1
Latin America	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.8	0.8	0.8
NGL & other liquids	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.3	5.3	5.4	5.4	5.4	5.3	5.4
Processing gains	2.6	2.6	2.5	2.6	2.5	2.7	2.7	2.8	2.7	2.7	2.7	2.8	2.6	2.6	2.7	2.7	2.7
Total	101.0	101.4	101.4	100.6	100.6	100.6	100.2	100.4	102.1	102.2	102.1	101.9	101.3	100.6	100.9	102.1	101.2

Source: Rystad, J.P. Morgan Commodities Research

Table 20: Global liquids supply, 2024F

mbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	65.5	65.5	65.6	65.7	66.1	66.6	66.8	66.6	66.5	67.0	67.0	66.7	65.5	66.1	66.7	66.9	66.3
Africa	1.2	1.2	1.2	1.2	1.2	1.2	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Asia	6.6	6.6	6.7	6.6	6.7	6.6	6.7	6.7	6.7	6.7	6.7	6.6	6.6	6.6	6.7	6.7	6.7
China	4.2	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
FSU	13.8	13.8	13.8	13.9	13.9	13.9	13.8	13.6	13.7	13.9	13.9	13.9	13.8	13.9	13.7	13.9	13.8
Russia	10.9	10.9	10.9	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	10.9	11.0	11.0	11.0	11.0
Latin America	6.3	6.3	6.3	6.3	6.4	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.3	6.4	6.5	6.5	6.4
Brazil	3.6	3.6	3.5	3.5	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6
Middle East	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Europe	3.2	3.2	3.2	3.2	3.1	3.1	3.1	3.0	2.9	3.1	3.2	3.2	3.2	3.1	3.0	3.1	3.1
Norway	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	1.8	2.0	2.0	2.0	2.0	2.0	1.9	2.0	2.0
North America	27.9	27.9	27.9	27.5	27.6	27.8	28.0	28.1	28.1	28.4	28.6	28.7	27.9	27.7	28.1	28.5	28.1
United States	19.9	19.9	19.8	19.9	19.9	20.0	20.0	20.1	20.2	20.3	20.4	20.5	19.9	19.9	20.1	20.4	20.1
Canada	6.0	6.0	6.0	5.6	5.6	5.8	5.9	5.9	5.8	6.0	6.1	6.1	6.0	5.7	5.9	6.1	5.9
OECD Pacific	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Other Liquids	2.9	2.9	3.0	3.3	3.7	3.9	3.9	3.9	3.8	3.7	3.5	3.1	2.9	3.6	3.9	3.4	3.5
OPEC	33.8	33.8	33.7	34.1	34.1	34.1	34.2	34.2	34.2	34.2	34.2	34.2	33.8	34.1	34.2	34.2	34.1
East of Suez	22.4	22.4	22.4	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.8	22.4	22.8	22.8	22.8	22.7
Africa	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Latin America	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.9	0.9	0.9	0.9
NGL & other liquids	5.5	5.5	5.4	5.4	5.5	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.5	5.4	5.4	5.4	5.4
Processing gains	2.7	2.6	2.6	2.6	2.6	2.7	2.7	2.7	2.7	2.6	2.7	2.7	2.6	2.6	2.7	2.7	2.7
Total	102.0	101.9	102.0	102.4	102.8	103.4	103.7	103.5	103.4	103.8	103.9	103.6	102.0	102.9	103.5	103.8	103.0

Source: Rystad, J.P. Morgan Commodities Research

Table 21: Global liquids supply, 2025F

mbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
Non-OPEC	67.1	67.3	67.4	67.6	67.9	68.3	68.7	68.6	68.3	68.8	68.8	68.5	67.3	67.9	68.5	68.7	68.1
Africa	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Asia	6.7	6.7	6.7	6.7	6.7	6.7	6.7	6.7	6.7	6.6	6.6	6.6	6.7	6.7	6.7	6.6	6.7
China	4.3	4.3	4.4	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
FSU	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9	13.9
Russia	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0
Latin America	6.6	6.6	6.7	6.7	6.8	6.9	7.0	7.0	7.1	7.2	7.2	7.3	6.6	6.8	7.0	7.2	6.9
Brazil	3.7	3.7	3.8	3.8	3.9	4.0	4.0	4.1	4.1	4.2	4.3	4.3	3.7	3.9	4.1	4.3	4.0
Middle East	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Europe	3.3	3.3	3.4	3.4	3.3	3.2	3.4	3.2	3.0	3.3	3.4	3.4	3.3	3.3	3.2	3.4	3.3
Norway	2.1	2.1	2.2	2.2	2.1	2.1	2.3	2.2	2.0	2.3	2.3	2.3	2.1	2.1	2.2	2.3	2.2
North America	28.9	29.0	28.9	28.6	28.7	28.8	28.8	28.9	28.9	29.2	29.3	29.3	28.9	28.7	28.9	29.2	28.9
United States	20.6	20.7	20.7	20.7	20.7	20.8	20.8	20.8	20.9	21.0	21.0	21.0	20.6	20.7	20.9	21.0	20.8
Canada	6.2	6.3	6.2	5.9	5.9	6.0	6.0	6.0	5.9	6.2	6.2	6.2	6.2	5.9	6.0	6.2	6.1
OECD Pacific	0.4	0.3	0.4	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.4	0.4	0.4	0.4	0.4
Other Liquids	3.0	3.0	3.1	3.4	3.8	3.9	4.0	4.0	3.9	3.8	3.6	3.2	3.0	3.7	4.0	3.5	3.6
OPEC	34.0	33.9	33.9	33.9	33.9	33.9	34.0	34.0	33.9	34.0	33.9	33.9	33.9	33.9	34.0	33.9	33.9
East of Suez	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6	22.6
Africa	4.8	4.8	4.8	4.8	4.8	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.8	4.7	4.7	4.7	4.7
Latin America	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
NGL & other liquids	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6
Processing gains	2.9	2.9	2.9	2.9	2.9	2.9	2.9	3.0	2.9	2.9	3.0	3.0	2.9	2.9	2.9	2.9	2.9
Total	104.0	104.1	104.2	104.4	104.8	105.1	105.6	105.5	105.2	105.7	105.7	105.4	104.1	104.7	105.4	105.6	105.0

Source: Rystad, J.P. Morgan Commodities Research

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Table 22: US crude and condensate production table, 2022E

	Jan-22	Feb-22	Mar-22	Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22	Nov-22	Dec-22
US crude and condensate production (kbd)	11,480	11,258	11,806	11,770	11,734	11,800	11,834	11,985	12,325	12,378	12,376	12,138
inclusive of												
Offshore Gulf of Mexico	1,680	1,613	1,685	1,754	1,607	1,735	1,727	1,761	1,824	1,789	1,797	1,788
Alaska	450	450	440	442	447	419	432	413	430	435	445	447
US tight oil basin crude and condensate production (kbd)	7,911	7,875	8,076	7,914	7,993	8,040	8,150	8,260	8,427	8,519	8,553	8,356
Anadarko	330	326	352	350	351	360	362	363	359	357	355	352
Bakken	1,123	1,138	1,158	962	1,090	1,115	1,103	1,111	1,142	1,150	1,151	1,028
Eagle Ford	1,081	1,058	1,074	1,089	1,088	1,094	1,092	1,105	1,126	1,139	1,125	1,125
Rockies	627	635	642	637	638	633	630	637	638	643	638	615
Permian Midland	2,154	2,108	2,171	2,157	2,157	2,198	2,264	2,320	2,371	2,403	2,435	2,403
Permian Delaware TX	1,261	1,232	1,241	1,234	1,198	1,157	1,179	1,181	1,191	1,189	1,190	1,177
Permian Delaware NM	1,336	1,378	1,438	1,485	1,471	1,484	1,519	1,543	1,600	1,638	1,659	1,656
US tight oil basin rig counts	443	467	482	506	525	540	550	555	550	557	569	572
Anadarko	48	50	48	48	52	53	55	58	57	59	64	67
Bakken	27	33	34	35	38	38	38	39	40	39	41	41
Eagle Ford	51	57	61	66	70	73	75	78	76	76	76	76
Rockies	25	25	26	28	29	31	32	34	34	37	40	38
Permian Midland	138	141	147	152	155	160	163	161	158	157	157	158
Permian Delaware TX	62	70	72	85	88	82	78	84	78	84	91	90
Permian Delaware NM	93	91	95	93	94	103	108	102	107	105	102	101
US tight oil basin active frac counts	141	155	157	160	169	175	176	173	175	178	179	177
Anadarko	12	13	10	12	14	11	9	9	10	10	10	11
Bakken	7	10	9	10	12	15	14	16	16	16	16	13
Eagle Ford	19	22	26	27	23	25	31	28	25	29	31	25
Rockies	10	10	14	12	14	13	12	12	14	14	16	17
Permian Midland	42	48	47	48	60	55	54	49	54	53	52	60
Permian Delaware TX	24	21	16	18	18	26	23	22	25	26	26	25
Permian Delaware NM	28	30	34	32	29	29	33	37	31	31	29	25
US tight oil basin drilled, uncompleted (DUC) well inventory	2,885	2,912	2,895	2,920	2,946	2,943	3,001	3,051	3,043	3,003	3,080	3,137
Anadarko	89	100	104	101	109	119	135	166	177	203	229	250
Bakken	397	397	381	408	412	403	381	357	332	319	335	358
Eagle Ford	322	318	296	285	311	330	334	335	361	349	354	371
Rockies	457	479	468	472	461	465	484	523	506	499	549	586
Permian Midland	808	802	825	819	775	742	748	792	786	788	820	795
Permian Delaware TX	282	283	308	326	345	336	361	352	366	334	326	333
Permian Delaware NM	530	532	513	509	532	547	557	525	515	511	466	445

Source: EIA, Rystad, Baker Hughes, J.P. Morgan Commodities Research

Table 23: US crude and condensate production table, 2023E

	Jan-23	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23
US crude and condensate production (kbd)	12,568	12,532	12,770	12,650	12,694	12,894	12,959	13,053	13,108	13,112	13,164	13,171
inclusive of												
Offshore Gulf of Mexico	1,903	1,828	1,874	1,736	1,708	1,859	1,933	1,891	1,949	1,924	1,963	1,963
Alaska	448	446	435	434	430	423	397	396	420	423	428	428
US tight oil basin crude and condensate production (kbd)	8,642	8,736	8,825	8,862	8,918	8,993	9,127	9,180	9,216	9,243	9,253	9,262
Anadarko	350	349	348	348	348	349	349	349	348	345	341	338
Bakken	1,106	1,173	1,150	1,163	1,175	1,204	1,238	1,248	1,253	1,252	1,252	1,249
Eagle Ford	1,158	1,167	1,188	1,191	1,208	1,229	1,235	1,228	1,219	1,210	1,203	1,197
Rockies	631	628	644	651	661	677	690	695	698	699	700	700
Permian Midland	2,482	2,499	2,531	2,517	2,530	2,541	2,560	2,549	2,544	2,543	2,546	2,550
Permian Delaware TX	1,194	1,199	1,223	1,246	1,263	1,283	1,323	1,333	1,344	1,354	1,365	1,376
Permian Delaware NM	1,722	1,722	1,741	1,746	1,734	1,710	1,732	1,778	1,811	1,840	1,846	1,852
US tight oil basin rig counts	575	568	563	557	534	512	499	482	470	464	470	470
Anadarko	66	66	63	57	50	44	42	39	37	38	38	38
Bakken	41	41	41	40	37	35	35	34	32	33	33	33
Eagle Ford	78	78	77	73	66	61	58	55	53	55	56	56
Rockies	35	31	32	32	31	30	30	31	30	28	27	27
Permian Midland	160	155	153	154	149	146	141	134	136	130	132	132
Permian Delaware TX	94	92	91	98	93	90	83	80	80	80	82	82
Permian Delaware NM	101	105	105	104	107	106	111	110	102	102	102	102
US tight oil basin active frac counts	180	188	188	183	183	178	170	171	166	165	166	166
Anadarko	12	11	14	14	15	13	13	13	10	10	10	10
Bakken	14	17	15	17	17	17	18	17	15	15	14	14
Eagle Ford	30	34	29	26	26	26	20	21	21	21	21	21
Rockies	16	16	20	21	22	21	18	17	17	16	16	16
Permian Midland	53	53	53	50	50	48	48	49	49	49	50	50
Permian Delaware TX	26	23	32	29	26	26	25	26	26	26	27	27
Permian Delaware NM	29	34	26	26	26	27	28	28	28	28	28	28
US tight oil basin drilled, uncompleted (DUC) well inventory	3,250	3,266	3,280	3,301	3,347	3,382	3,421	3,422	3,427	3,410	3,412	3,399
Anadarko	259	267	272	274	266	257	244	225	220	214	209	204
Bakken	364	359	355	350	343	337	328	323	323	310	317	309
Eagle Ford	388	362	379	380	395	379	394	392	387	387	387	388
Rockies	627	646	653	625	616	606	616	617	616	613	609	605
Permian Midland	822	828	827	859	888	939	958	960	964	958	951	944
Permian Delaware TX	351	372	350	357	372	385	389	386	383	380	375	371
Permian Delaware NM	439	432	444	455	466	479	491	519	534	548	563	578

Source: EIA, Rystad, Baker Hughes, J.P. Morgan Commodities Research

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Table 24: US crude and condensate production table, 2024F

	Jan-24	Feb-24	Mar-24	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24
US crude and condensate production (kbd)	13,150	13,152	13,162	13,177	13,197	13,212	13,245	13,313	13,352	13,418	13,507	13,549
inclusive of												
Offshore Gulf of Mexico	1,946	1,931	1,929	1,923	1,929	1,915	1,927	1,923	1,886	1,897	1,935	1,940
Alaska	419	424	425	426	409	396	373	398	419	424	429	428
US tight oil basin crude and condensate production (kbd)	9,267	9,281	9,293	9,316	9,349	9,393	9,438	9,487	9,543	9,593	9,640	9,680
Anadarko	337	337	336	336	335	334	334	333	333	333	332	332
Bakken	1,251	1,257	1,261	1,265	1,268	1,272	1,274	1,277	1,280	1,281	1,282	1,282
Eagle Ford	1,191	1,188	1,184	1,181	1,179	1,179	1,180	1,181	1,183	1,184	1,185	1,185
Rockies	700	702	704	707	710	712	715	717	720	722	724	725
Permian Midland	2,553	2,560	2,568	2,577	2,586	2,598	2,610	2,623	2,637	2,650	2,662	2,673
Permian Delaware TX	1,383	1,389	1,392	1,394	1,398	1,402	1,407	1,415	1,427	1,439	1,450	1,460
Permian Delaware NM	1,851	1,849	1,849	1,857	1,874	1,896	1,918	1,941	1,964	1,985	2,005	2,023
US tight oil basin rig counts	480	482	487	491	495	496	510	510	510	510	510	510
Anadarko	40	40	40	40	40	40	40	40	40	40	40	40
Bakken	33	34	35	35	35	35	35	35	35	35	35	35
Eagle Ford	57	57	58	58	60	60	60	60	60	60	60	60
Rockies	27	28	28	29	29	30	30	30	30	30	30	30
Permian Midland	136	136	136	139	139	139	145	145	145	145	145	145
Permian Delaware TX	85	85	85	85	85	85	90	90	90	90	90	90
Permian Delaware NM	102	102	105	105	107	107	110	110	110	110	110	110
US tight oil basin active frac counts	169	169	174	175	177	177	182	182	182	182	182	182
Anadarko	10	10	10	10	10	10	10	10	10	10	10	10
Bakken	14	14	14	14	14	14	14	14	14	14	14	14
Eagle Ford	23	23	23	23	24	24	24	24	24	24	24	24
Rockies	15	15	15	15	15	15	15	15	15	15	15	15
Permian Midland	51	51	51	52	52	52	53	53	53	53	53	53
Permian Delaware TX	29	29	29	29	29	29	32	32	32	32	32	32
Permian Delaware NM	28	28	33	33	34	34	35	35	35	35	35	35
US tight oil basin drilled, uncompleted (DUC) well inventory	3,408	3,422	3,422	3,424	3,424	3,427	3,431	3,435	3,439	3,442	3,446	3,450
Anadarko	205	205	205	205	205	205	205	205	205	205	205	205
Bakken	304	301	300	300	299	298	298	297	296	296	295	294
Eagle Ford	388	387	389	390	391	391	392	392	392	393	393	394
Rockies	595	589	583	580	577	577	577	577	577	577	578	578
Permian Midland	937	931	924	918	911	905	905	905	905	904	904	904
Permian Delaware TX	379	387	395	404	412	420	423	427	430	433	436	440
Permian Delaware NM	602	624	626	629	630	631	632	632	633	634	634	635

Source: EIA, Rystad, Baker Hughes, J.P. Morgan Commodities Research

Table 25: US crude and condensate production table, 2025F

	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25
US crude and condensate production (kbd)	13,580	13,643	13,652	13,658	13,663	13,667	13,653	13,694	13,779	13,790	13,790	13,788
inclusive of												
Offshore Gulf of Mexico	1,951	1,998	2,005	2,008	2,011	2,020	2,007	2,023	2,048	2,025	1,999	1,976
Alaska	431	437	440	439	427	401	378	379	412	422	426	430
US tight oil basin crude and condensate production (kbd)	9,699	9,712	9,715	9,721	9,735	9,758	9,780	9,805	9,832	9,855	9,877	9,896
Anadarko	331	330	329	328	328	327	327	327	327	326	326	325
Bakken	1,286	1,289	1,288	1,286	1,286	1,287	1,287	1,287	1,288	1,287	1,287	1,286
Eagle Ford	1,184	1,183	1,180	1,178	1,177	1,177	1,177	1,177	1,178	1,178	1,178	1,178
Rockies	726	726	724	722	721	720	719	719	719	719	719	719
Permian Midland	2,675	2,676	2,676	2,678	2,681	2,687	2,694	2,701	2,708	2,716	2,722	2,728
Permian Delaware TX	1,466	1,471	1,477	1,483	1,491	1,499	1,507	1,515	1,523	1,530	1,537	1,544
Permian Delaware NM	2,031	2,037	2,040	2,045	2,052	2,061	2,070	2,080	2,090	2,099	2,108	2,117
US tight oil basin rig counts	508	508	508	508	508	508	508	508	508	508	508	508
Anadarko	39	39	39	39	39	39	39	39	39	39	39	39
Bakken	34	34	34	34	34	34	34	34	34	34	34	34
Eagle Ford	59	59	59	59	59	59	59	59	59	59	59	59
Rockies	28	28	28	28	28	28	28	28	28	28	28	28
Permian Midland	148	148	148	148	148	148	148	148	148	148	148	148
Permian Delaware TX	92	92	92	92	92	92	92	92	92	92	92	92
Permian Delaware NM	107	107	107	107	107	107	107	107	107	107	107	107
US tight oil basin active frac counts	181	181	181	181	181	181	181	181	181	181	181	181
Anadarko	9	9	9	9	9	9	9	9	9	9	9	9
Bakken	13	13	13	13	13	13	13	13	13	13	13	13
Eagle Ford	23	23	23	23	23	23	23	23	23	23	23	23
Rockies	14	14	14	14	14	14	14	14	14	14	14	14
Permian Midland	54	54	54	54	54	54	54	54	54	54	54	54
Permian Delaware TX	33	33	33	33	33	33	33	33	33	33	33	33
Permian Delaware NM	34	34	34	34	34	34	34	34	34	34	34	34
US tight oil basin drilled, uncompleted (DUC) well inventory	3,454	3,457	3,461	3,465	3,469	3,473	3,477	3,481	3,485	3,489	3,493	3,496
Anadarko	205	205	205	205	205	205	205	205	205	205	205	205
Bakken	297	299	302	305	307	310	313	315	318	320	323	326
Eagle Ford	394	394	394	395	395	395	396	396	396	396	397	397
Rockies	578	578	578	579	579	579	579	580	580	580	580	580
Permian Midland	904	904	905	905	905	905	905	906	906	906	906	906
Permian Delaware TX	440	440	440	440	440	440	441	441	441	441	441	441
Permian Delaware NM	636	636	637	637	638	638	639	639	640	640	641	641

Source: EIA, Rystad, Baker Hughes, J.P. Morgan Commodities Research

Table 26: Regional oil demand, 2022E

mbd	2022E												2022E				2022E
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
North America	23.6	24.4	24.6	24.0	24.1	25.0	24.5	24.6	24.5	24.3	24.7	23.8	24.2	24.4	24.5	24.3	24.3
United States	19.6	20.2	20.5	19.7	19.8	20.4	19.9	20.3	20.1	20.0	20.2	19.3	20.1	20.0	20.1	19.8	20.0
Canada	2.4	2.5	2.2	2.3	2.3	2.5	2.5	2.4	2.4	2.4	2.5	2.5	2.4	2.4	2.4	2.5	2.4
Mexico	1.6	1.7	1.9	2.0	2.0	2.0	2.0	1.9	1.9	2.0	2.0	2.0	1.8	2.0	2.0	2.0	1.9
Latin America	6.2	6.5	6.5	6.6	6.6	6.4	6.6	6.8	6.8	6.5	6.6	6.8	6.4	6.5	6.7	6.7	6.6
Brazil	2.9	3.0	3.1	3.0	3.1	2.9	3.1	3.3	3.1	3.1	3.2	3.3	3.0	3.0	3.2	3.2	3.1
Other Latin America	3.3	3.5	3.4	3.5	3.5	3.5	3.4	3.6	3.6	3.5	3.5	3.6	3.4	3.5	3.5	3.5	3.5
Europe	13.0	14.6	14.3	14.0	14.2	14.6	14.5	14.9	15.0	14.0	14.2	14.2	14.0	14.2	14.8	14.1	14.3
FSU	4.7	4.9	4.8	4.8	4.6	4.9	5.1	5.1	5.2	5.1	5.1	5.1	4.8	4.8	5.1	5.1	4.9
Middle East	8.6	8.7	8.7	9.1	9.1	9.5	9.5	9.6	9.6	9.1	9.1	9.0	8.7	9.2	9.6	9.1	9.1
Africa	4.1	4.2	4.4	4.3	4.2	4.2	4.2	4.2	4.3	4.4	4.3	4.4	4.3	4.2	4.2	4.4	4.3
Asia Pacific	37.4	37.5	35.7	34.4	34.9	35.5	35.0	35.4	34.7	36.1	36.5	37.4	36.9	34.9	35.0	36.7	35.9
Australia	1.0	1.1	1.1	1.0	1.1	1.1	1.1	1.1	1.1	1.0	1.2	1.1	1.0	1.1	1.1	1.1	1.1
China	15.8	15.6	14.4	13.7	14.3	14.6	14.7	14.8	14.6	15.5	15.1	15.0	15.3	14.2	14.7	15.2	14.9
India	5.1	5.5	5.5	5.4	5.1	5.4	5.0	5.0	5.0	5.2	5.5	5.5	5.4	5.3	5.0	5.4	5.3
Japan	3.8	3.8	3.5	3.0	3.0	3.1	3.1	3.3	3.2	3.2	3.5	4.0	3.7	3.0	3.2	3.6	3.4
Korea	2.8	2.7	2.6	2.5	2.6	2.3	2.5	2.6	2.4	2.4	2.5	2.8	2.7	2.5	2.5	2.5	2.5
Other Asia Pacific	8.9	8.8	8.7	8.8	8.8	8.9	8.7	8.5	8.4	8.6	8.8	8.9	8.8	8.8	8.5	8.8	8.7
World oil demand	97.8	100.8	99.0	97.0	97.8	100.1	99.4	100.7	100.1	99.5	100.5	100.9	99.1	98.3	100.1	100.3	99.4

Source: Wood Mackenzie, IEA, FlightAware, EIA, MariTrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

Table 27: Regional oil demand, 2023E

mbd	2023F												2023F				2023F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
North America	23.3	24.0	24.3	24.1	24.8	25.3	24.8	25.6	24.8	24.9	24.8	24.5	23.9	24.7	25.0	24.7	24.6
United States	19.1	19.8	20.1	20.0	20.4	20.7	20.1	20.9	20.4	20.4	20.4	20.1	19.7	20.4	20.5	20.3	20.2
Canada	2.3	2.4	2.3	2.3	2.5	2.6	2.7	2.7	2.5	2.5	2.5	2.5	2.3	2.5	2.6	2.5	2.5
Mexico	1.9	1.9	1.9	1.8	1.9	1.9	1.9	2.0	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Latin America	6.4	6.6	6.7	6.6	6.8	6.7	6.8	6.9	6.7	6.8	6.7	6.6	6.6	6.7	6.8	6.7	6.7
Brazil	3.0	3.1	3.2	3.1	3.2	3.2	3.3	3.3	3.2	3.3	3.2	3.1	3.1	3.2	3.3	3.2	3.2
Other Latin America	3.4	3.5	3.5	3.5	3.6	3.5	3.5	3.6	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Europe	13.0	14.6	14.1	13.8	14.4	14.8	14.3	14.3	14.4	14.6	14.3	13.9	13.9	14.3	14.3	14.3	14.2
FSU	4.7	5.0	4.9	4.9	4.8	5.0	5.1	5.3	5.1	5.0	5.0	5.2	4.9	4.9	5.2	5.1	5.0
Middle East	9.0	9.0	9.0	8.9	9.1	9.3	9.6	9.8	9.7	9.3	9.2	8.9	9.0	9.1	9.7	9.1	9.2
Africa	4.2	4.3	4.4	4.2	4.2	4.1	4.1	4.2	4.0	4.1	4.1	4.5	4.3	4.2	4.1	4.2	4.2
Asia Pacific	37.0	37.6	36.9	37.0	37.1	37.2	37.3	37.5	36.9	37.3	38.1	39.1	37.1	37.1	37.2	38.1	37.4
Australia	1.0	1.1	1.2	1.1	1.2	1.1	1.1	1.1	1.1	1.1	1.2	1.1	1.1	1.1	1.1	1.2	1.1
China	15.4	15.3	15.1	16.2	16.2	16.1	16.7	16.8	16.0	16.5	16.2	16.7	15.3	16.2	16.5	16.5	16.1
India	5.3	5.7	5.7	5.3	5.6	5.6	5.1	5.3	5.3	5.3	5.8	5.7	5.6	5.5	5.2	5.6	5.5
Japan	3.8	3.9	3.5	3.2	3.0	3.1	3.1	3.1	3.1	3.1	3.4	3.8	3.7	3.1	3.1	3.4	3.3
Korea	2.6	2.6	2.7	2.4	2.4	2.4	2.5	2.5	2.5	2.5	2.7	2.8	2.6	2.4	2.5	2.7	2.5
Other Asia Pacific	8.8	8.8	8.7	8.8	8.7	8.9	8.8	8.7	8.9	8.6	8.8	8.9	8.8	8.8	8.8	8.8	8.8
World oil demand	97.6	101.1	100.3	99.5	101.2	102.2	102.1	103.4	101.7	102.0	102.2	102.7	99.6	101.0	102.4	102.3	101.3

Source: Wood Mackenzie, IEA, FlightAware, EIA, MariTrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

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Table 28: Regional oil demand, 2024F

mbd	2024F												2024F				2024F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
North America	24.7	24.5	24.3	24.5	24.6	25.1	25.1	25.9	24.5	25.0	24.8	24.6	24.5	24.7	25.2	24.8	24.8
United States	20.3	20.0	20.0	20.1	20.2	20.6	20.5	21.2	20.1	20.6	20.5	20.2	20.1	20.3	20.6	20.4	20.4
Canada	2.5	2.5	2.4	2.2	2.4	2.5	2.5	2.6	2.5	2.5	2.5	2.5	2.5	2.3	2.5	2.5	2.5
Mexico	1.9	2.0	1.9	2.2	2.0	2.1	2.1	2.0	1.9	1.9	1.9	1.9	1.9	2.1	2.0	1.9	2.0
Latin America	6.4	6.8	6.7	6.7	6.8	6.7	7.0	6.9	6.8	7.0	6.8	6.7	6.6	6.8	6.9	6.8	6.8
Brazil	2.9	3.3	3.2	3.1	3.2	3.1	3.4	3.3	3.3	3.4	3.3	3.2	3.1	3.2	3.3	3.3	3.2
Other Latin America	3.5	3.5	3.5	3.6	3.6	3.5	3.6	3.6	3.5	3.5	3.6	3.5	3.5	3.6	3.6	3.5	3.5
Europe	13.4	14.4	14.1	14.1	14.2	14.6	14.8	14.6	14.5	14.5	14.2	13.9	14.0	14.3	14.6	14.2	14.3
FSU	4.8	4.9	4.8	4.9	4.9	5.0	5.2	5.4	5.2	5.2	5.1	5.3	4.8	5.0	5.3	5.2	5.1
Middle East	9.0	9.2	9.0	9.1	9.1	9.3	9.6	9.8	9.8	9.4	9.3	9.0	9.1	9.2	9.8	9.3	9.3
Africa	4.3	4.3	4.4	4.4	4.4	4.3	4.3	4.3	4.1	4.1	4.2	4.5	4.4	4.4	4.2	4.3	4.3
Asia Pacific	38.0	38.5	37.9	38.3	38.0	38.0	38.4	38.3	37.5	38.2	39.4	40.0	38.1	38.1	38.1	39.2	38.4
Australia	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.1	1.1	1.2	1.2	1.2	1.2
China	15.8	15.8	15.7	16.4	16.4	16.4	17.1	16.9	16.4	17.4	17.4	17.4	15.8	16.4	16.8	17.4	16.6
India	5.7	6.0	6.0	5.9	5.9	5.7	5.6	5.6	5.4	5.5	6.0	5.9	5.9	5.8	5.5	5.8	5.8
Japan	3.6	3.8	3.4	3.3	3.0	3.0	3.1	3.1	3.2	3.1	3.4	3.7	3.6	3.1	3.1	3.4	3.3
Korea	2.8	2.7	2.6	2.6	2.5	2.5	2.6	2.7	2.6	2.6	2.7	2.9	2.7	2.5	2.6	2.7	2.7
Other Asia Pacific	9.0	9.0	9.1	9.0	9.0	9.1	8.8	8.8	8.9	8.5	8.7	8.9	9.0	9.0	8.8	8.7	8.9
World oil demand	100.7	102.6	101.3	102.1	102.1	102.9	104.4	105.1	102.4	103.4	103.9	104.2	101.5	102.4	104.0	103.8	102.9

Source: Wood Mackenzie, IEA, FlightAware, EIA, MariTrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

Table 29: Regional oil demand, 2025F

mbd	2025F												2025F				2025F
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
North America	24.7	24.4	24.1	24.3	24.4	25.0	25.2	25.6	24.5	25.0	24.8	24.5	24.4	24.6	25.1	24.8	24.7
United States	20.2	19.9	19.8	19.9	19.9	20.4	20.5	21.0	20.1	20.6	20.5	20.1	20.0	20.1	20.5	20.4	20.2
Canada	2.5	2.6	2.4	2.2	2.4	2.5	2.5	2.6	2.5	2.5	2.5	2.5	2.5	2.4	2.6	2.5	2.5
Mexico	1.9	2.0	1.9	2.2	2.0	2.1	2.1	2.0	1.9	1.9	1.9	1.9	1.9	2.1	2.0	1.9	2.0
Latin America	6.5	6.9	6.6	6.8	6.8	6.7	7.1	6.9	6.8	7.0	6.9	6.7	6.7	6.8	6.9	6.9	6.8
Brazil	3.0	3.4	3.2	3.3	3.3	3.2	3.5	3.4	3.3	3.5	3.4	3.2	3.2	3.3	3.4	3.4	3.3
Other Latin America	3.5	3.5	3.4	3.5	3.6	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.4	3.5	3.5	3.5	3.5
Europe	13.8	14.2	13.8	14.3	14.0	14.3	15.0	14.6	14.4	14.4	14.1	13.8	13.9	14.2	14.7	14.1	14.2
FSU	4.8	5.0	4.9	5.0	5.0	5.1	5.3	5.5	5.3	5.3	5.2	5.4	4.9	5.1	5.4	5.3	5.2
Middle East	9.1	9.2	9.1	9.2	9.2	9.4	9.7	9.9	9.9	9.5	9.3	9.1	9.1	9.3	9.9	9.3	9.4
Africa	4.5	4.4	4.5	4.5	4.5	4.4	4.4	4.4	4.2	4.2	4.3	4.6	4.5	4.5	4.3	4.4	4.4
Asia Pacific	38.9	39.4	38.3	39.3	39.1	39.2	39.3	39.2	38.1	38.9	40.0	40.5	38.8	39.2	38.9	39.8	39.2
Australia	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.1	1.2	1.2	1.2	1.2
China	16.4	16.5	15.8	17.0	17.0	17.0	17.4	17.2	16.6	17.6	17.6	17.6	16.2	17.0	17.0	17.6	17.0
India	6.0	6.2	6.3	6.1	6.2	6.0	5.9	5.8	5.6	5.8	6.3	6.2	6.1	6.1	5.8	6.1	6.0
Japan	3.5	3.7	3.3	3.2	3.0	3.0	3.1	3.2	3.2	3.1	3.3	3.6	3.5	3.1	3.1	3.3	3.3
Korea	2.8	2.7	2.6	2.7	2.5	2.6	2.7	2.9	2.6	2.6	2.8	2.9	2.7	2.6	2.7	2.8	2.7
Other Asia Pacific	9.1	9.1	9.1	9.2	9.2	9.4	9.0	9.0	9.1	8.7	8.8	9.0	9.1	9.2	9.0	8.8	9.0
World oil demand	102.2	103.6	101.3	103.4	103.1	104.0	106.0	106.1	103.2	104.3	104.6	104.7	102.3	103.5	105.1	104.5	103.9

Source: Wood Mackenzie, IEA, FlightAware, EIA, MariTrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

Table 30: Regional oil demand by product, 2022E

mbd		2022E												2022E				2022E
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
United States																		
	LPG	1.5	1.4	1.2	0.9	0.8	0.9	0.9	0.9	1.0	1.0	1.2	1.3	1.4	0.8	0.9	1.1	1.1
	Naphtha	0.2	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1
	Gasoline	8.1	8.7	9.0	8.8	9.1	9.1	8.8	9.1	8.8	8.8	8.8	8.6	8.6	9.0	8.9	8.7	8.8
	Jet and Kerosene	1.4	1.4	1.5	1.5	1.6	1.7	1.6	1.7	1.5	1.6	1.6	1.6	1.5	1.6	1.6	1.6	1.6
	Diesel/Gasoil	4.1	4.4	4.2	4.0	3.9	4.0	3.7	3.9	4.1	4.2	4.1	3.8	4.2	4.0	3.9	4.0	4.0
	Fuel Oil	0.3	0.3	0.4	0.3	0.3	0.3	0.3	0.4	0.5	0.3	0.4	0.3	0.3	0.3	0.4	0.3	0.3
	Other Products	4.0	3.9	4.0	4.1	4.0	4.3	4.5	4.2	4.1	4.1	4.0	3.7	4.0	4.2	4.2	3.9	4.1
	Total Oil	19.6	20.2	20.5	19.7	19.8	20.4	19.9	20.3	20.1	20.0	20.2	19.3	20.1	20.0	20.1	19.8	20.0
Europe																		
	LPG	1.0	1.1	1.1	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.9	1.0	1.1	0.9	1.0	1.0	1.0
	Naphtha	1.1	1.1	0.9	0.9	0.9	0.9	0.8	0.8	0.7	0.7	0.8	0.8	1.1	0.9	0.8	0.8	0.9
	Gasoline	1.9	2.1	2.1	2.1	2.2	2.4	2.3	2.4	2.2	2.1	2.2	2.2	2.0	2.2	2.3	2.2	2.2
	Jet and Kerosene	1.1	1.2	1.2	1.4	1.4	1.6	1.6	1.7	1.7	1.6	1.4	1.5	1.2	1.5	1.7	1.5	1.4
	Diesel/Gasoil	5.9	6.9	6.8	6.5	6.4	6.5	6.5	6.7	7.0	6.5	6.7	6.7	6.5	6.5	6.7	6.6	6.6
	Fuel Oil	0.8	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.9	0.8	0.9	0.9	0.8	0.9
	Other Products	1.3	1.3	1.4	1.3	1.4	1.4	1.4	1.5	1.5	1.3	1.3	1.2	1.3	1.4	1.5	1.3	1.4
	Total Oil	13.0	14.6	14.3	14.0	14.2	14.6	14.5	14.9	15.0	14.0	14.2	14.2	14.0	14.2	14.8	14.1	14.3
China																		
	LPG	2.3	2.3	2.2	2.1	2.2	2.3	2.2	2.2	2.3	2.4	2.5	2.5	2.3	2.2	2.3	2.5	2.3
	Naphtha	1.8	1.8	1.6	1.5	1.5	1.6	1.6	1.6	1.6	1.8	1.8	1.8	1.7	1.5	1.6	1.8	1.7
	Gasoline	4.0	4.2	3.3	3.1	3.5	3.6	3.6	3.7	3.5	3.7	3.4	3.2	3.8	3.4	3.6	3.4	3.6
	Jet and Kerosene	0.6	0.6	0.5	0.3	0.3	0.5	0.6	0.6	0.4	0.4	0.4	0.5	0.6	0.4	0.5	0.4	0.5
	Diesel/Gasoil	4.1	3.7	3.7	3.8	4.0	3.8	3.8	3.9	3.9	4.3	4.1	4.0	3.9	3.9	3.9	4.1	3.9
	Fuel Oil	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.5	0.6	0.6	0.6	0.6	0.6
	Other Products	2.4	2.4	2.4	2.3	2.3	2.3	2.3	2.3	2.3	2.4	2.4	2.4	2.4	2.3	2.3	2.4	2.3
	Total Oil	15.8	15.6	14.4	13.7	14.3	14.6	14.7	14.8	14.6	15.5	15.1	15.0	15.3	14.2	14.7	15.2	14.9
World																		
	LPG	10.7	10.8	10.3	9.5	9.4	9.6	9.8	9.8	9.9	9.8	10.4	10.8	10.6	9.5	9.8	10.3	10.1
	Naphtha	7.1	6.8	6.5	6.3	6.2	6.2	6.4	6.3	6.1	6.0	6.5	6.6	6.8	6.2	6.3	6.4	6.4
	Gasoline	24.7	25.9	25.5	25.2	26.1	26.5	26.3	26.9	26.2	26.1	26.0	26.1	25.4	26.0	26.4	26.0	26.0
	Jet and Kerosene	6.3	6.3	6.0	5.8	6.0	6.5	6.5	6.6	6.5	6.5	6.4	7.0	6.2	6.1	6.5	6.6	6.4
	Diesel/Gasoil	27.8	29.7	29.4	28.8	28.7	29.3	28.4	29.2	29.5	29.7	29.8	29.2	29.0	28.9	29.0	29.6	29.1
	Fuel Oil	6.4	6.4	6.5	6.4	6.5	6.8	6.8	6.8	7.0	6.5	6.7	6.6	6.4	6.6	6.9	6.6	6.6
	Other Products	14.7	14.8	14.8	14.9	14.9	15.3	15.3	15.0	15.0	14.8	14.9	14.5	14.8	15.0	15.1	14.7	14.9
	Total Oil	97.8	100.8	99.0	97.0	97.8	100.1	99.4	100.7	100.1	99.5	100.5	100.9	99.1	98.3	100.1	100.3	99.4

Source: Wood Mackenzie, IEA, FlightAware, EIA, MarITrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

Table 31: Regional oil demand by product, 2023E

mbd		2023F												2023F				2023F
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
United States																		
	LPG	1.3	1.1	1.0	0.9	0.8	0.9	0.9	0.9	1.0	1.1	1.2	1.2	1.2	0.9	0.9	1.2	1.0
	Naphtha	0.1	0.1	0.1	0.2	0.2	0.1	0.1	0.1	0.2	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.1
	Gasoline	8.3	8.7	9.0	9.0	9.1	9.3	9.0	9.3	9.0	8.9	8.8	8.6	8.7	9.1	9.1	8.8	8.9
	Jet and Kerosene	1.5	1.5	1.6	1.6	1.7	1.7	1.8	1.7	1.6	1.7	1.6	1.7	1.5	1.7	1.7	1.7	1.6
	Diesel/Gasoil	3.9	4.0	4.1	3.9	3.9	4.0	3.6	4.1	4.0	4.1	4.1	3.8	4.0	3.9	3.9	4.0	4.0
	Fuel Oil	0.3	0.4	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.2	0.3	0.3	0.2	0.3	0.3	0.3
	Other Products	3.7	3.9	3.9	4.3	4.5	4.4	4.4	4.4	4.3	4.2	4.4	4.3	3.9	4.4	4.3	4.3	4.2
	Total Oil	19.1	19.8	20.1	20.0	20.4	20.7	20.1	20.9	20.4	20.4	20.4	20.1	19.7	20.4	20.5	20.3	20.2
Europe																		
	LPG	1.1	1.1	1.0	1.0	1.0	1.0	1.0	0.9	1.0	1.0	1.0	1.1	1.1	1.0	1.0	1.0	1.0
	Naphtha	0.9	0.9	0.9	0.9	0.7	0.7	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.8	0.7	0.9	0.8
	Gasoline	2.0	2.3	2.2	2.2	2.4	2.4	2.4	2.4	2.2	2.3	2.2	2.2	2.1	2.3	2.4	2.2	2.3
	Jet and Kerosene	1.4	1.5	1.4	1.5	1.6	1.7	1.8	1.8	1.8	1.7	1.6	1.6	1.4	1.6	1.8	1.6	1.6
	Diesel/Gasoil	5.7	6.7	6.5	6.1	6.5	6.6	6.3	6.2	6.5	6.7	6.5	6.3	6.3	6.4	6.4	6.5	6.4
	Fuel Oil	0.8	0.9	0.8	0.8	0.8	0.8	0.9	0.8	0.8	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8
	Other Products	1.2	1.3	1.2	1.2	1.3	1.5	1.3	1.3	1.3	1.3	1.3	1.2	1.2	1.3	1.3	1.2	1.3
	Total Oil	13.0	14.6	14.1	13.8	14.4	14.8	14.3	14.3	14.4	14.6	14.3	13.9	13.9	14.3	14.3	14.3	14.2
China																		
	LPG	2.4	2.4	2.2	2.3	2.3	2.3	2.6	2.6	2.4	2.4	2.6	2.6	2.3	2.3	2.5	2.5	2.4
	Naphtha	1.8	1.8	1.8	1.9	1.9	1.8	1.8	1.8	1.9	1.9	2.0	1.9	1.8	1.8	1.8	1.9	1.8
	Gasoline	3.8	3.8	3.6	4.0	4.0	4.0	4.0	4.2	4.0	4.0	3.8	3.9	3.7	4.0	4.1	3.9	3.9
	Jet and Kerosene	0.6	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.7	0.8	0.7	0.8	0.6	0.7	0.8	0.8	0.7
	Diesel/Gasoil	3.9	3.8	3.9	4.2	4.2	4.2	4.4	4.4	4.2	4.2	4.1	4.3	3.9	4.2	4.3	4.2	4.2
	Fuel Oil	0.5	0.6	0.6	0.7	0.7	0.7	0.6	0.4	0.4	0.7	0.6	0.5	0.6	0.7	0.5	0.6	0.6
	Other Products	2.4	2.4	2.3	2.4	2.4	2.4	2.5	2.6	2.4	2.6	2.3	2.6	2.4	2.4	2.5	2.5	2.4
	Total Oil	15.4	15.3	15.1	16.2	16.2	16.1	16.7	16.8	16.0	16.5	16.2	16.7	15.3	16.2	16.5	16.5	16.1
World																		
	LPG	10.7	10.6	10.0	9.8	9.8	9.9	10.2	10.2	10.1	10.3	10.6	10.8	10.5	9.8	10.2	10.5	10.2
	Naphtha	6.8	6.6	6.6	6.4	6.2	6.1	6.2	6.2	6.4	6.3	6.8	6.8	6.7	6.3	6.3	6.6	6.5
	Gasoline	25.1	26.2	26.2	26.5	27.0	27.3	27.1	27.8	26.8	26.7	26.6	26.9	25.8	27.0	27.2	26.7	26.7
	Jet and Kerosene	7.1	7.2	6.9	7.0	7.1	7.5	7.5	7.6	7.4	7.5	7.3	7.7	7.1	7.2	7.5	7.5	7.3
	Diesel/Gasoil	27.1	29.2	29.3	28.3	29.4	29.4	29.0	29.7	29.2	29.9	29.6	29.0	28.5	29.1	29.3	29.5	29.1
	Fuel Oil	6.4	6.6	6.5	6.4	6.5	6.6	6.6	6.5	6.5	6.3	6.1	6.1	6.5	6.5	6.5	6.2	6.4
	Other Products	14.4	14.8	14.7	14.9	15.3	15.4	15.4	15.5	15.2	15.1	15.2	15.5	14.6	15.2	15.4	15.2	15.1
	Total Oil	97.6	101.1	100.3	99.5	101.2	102.2	102.1	103.4	101.7	102.0	102.2	102.7	99.6	101.0	102.4	102.3	101.3

Source: Wood Mackenzie, IEA, FlightAware, EIA, MarITrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

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Table 32: Regional oil demand by product, 2024F

mbd		2024F												2024F				2024F
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
United States																		
	LPG	1.3	1.2	1.0	0.9	0.9	0.9	0.9	0.9	1.0	1.1	1.2	1.2	1.2	0.9	0.9	1.2	1.0
	Naphtha	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.2	0.1	0.1
	Gasoline	8.4	8.6	8.8	8.9	9.0	9.2	9.0	9.4	8.7	8.9	8.8	8.5	8.6	9.0	9.0	8.8	8.8
	Jet and Kerosene	1.6	1.7	1.7	1.8	1.8	1.8	1.9	1.9	1.7	1.8	1.7	1.8	1.7	1.8	1.8	1.7	1.8
	Diesel/Gasoil	4.1	4.1	4.1	3.9	3.9	3.9	3.7	4.1	3.8	4.0	4.0	3.8	4.1	3.9	3.8	4.0	4.0
	Fuel Oil	0.4	0.3	0.3	0.2	0.2	0.4	0.4	0.4	0.3	0.4	0.2	0.3	0.3	0.3	0.4	0.3	0.3
	Other Products	4.4	4.0	4.1	4.3	4.3	4.3	4.4	4.5	4.4	4.3	4.5	4.4	4.1	4.3	4.4	4.4	4.3
	Total Oil	20.3	20.0	20.0	20.1	20.2	20.6	20.5	21.2	20.1	20.6	20.5	20.2	20.1	20.3	20.6	20.4	20.4
Europe																		
	LPG	1.1	1.1	1.0	1.1	1.0	1.0	1.1	1.1	1.0	1.0	1.0	1.1	1.0	1.1	1.1	1.0	1.1
	Naphtha	1.0	1.0	1.0	0.9	0.7	0.7	0.9	0.8	0.8	0.8	0.9	0.9	1.0	0.8	0.8	0.9	0.9
	Gasoline	2.1	2.2	2.1	2.3	2.3	2.3	2.4	2.4	2.3	2.3	2.2	2.2	2.1	2.3	2.3	2.2	2.3
	Jet and Kerosene	1.5	1.5	1.6	1.6	1.7	1.8	1.8	1.8	1.7	1.7	1.6	1.7	1.5	1.7	1.8	1.7	1.7
	Diesel/Gasoil	5.8	6.6	6.3	6.1	6.4	6.5	6.4	6.3	6.6	6.7	6.5	6.2	6.2	6.3	6.4	6.4	6.3
	Fuel Oil	0.8	0.9	0.8	0.8	0.9	0.9	0.9	0.8	0.7	0.8	0.7	0.8	0.8	0.9	0.8	0.7	0.8
	Other Products	1.2	1.2	1.3	1.3	1.3	1.3	1.4	1.3	1.3	1.2	1.2	1.1	1.2	1.3	1.3	1.2	1.3
	Total Oil	13.4	14.4	14.1	14.1	14.2	14.6	14.8	14.6	14.5	14.5	14.2	13.9	14.0	14.3	14.6	14.2	14.3
China																		
	LPG	2.4	2.4	2.4	2.3	2.3	2.3	2.6	2.5	2.4	2.5	2.7	2.7	2.4	2.3	2.5	2.6	2.5
	Naphtha	1.9	1.8	2.0	2.0	2.0	2.0	2.0	2.0	1.9	2.1	2.2	2.3	1.9	2.0	2.0	2.2	2.0
	Gasoline	3.8	3.9	3.7	3.9	3.9	4.0	4.2	4.1	3.9	4.1	4.0	4.0	3.8	3.9	4.1	4.0	4.0
	Jet and Kerosene	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.9	1.0	0.9	0.8	0.8	0.8	0.8	0.9	0.8
	Diesel/Gasoil	4.0	4.0	4.0	4.3	4.3	4.3	4.4	4.4	4.3	4.4	4.4	4.5	4.0	4.3	4.4	4.4	4.3
	Fuel Oil	0.6	0.7	0.7	0.7	0.6	0.7	0.6	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.6	0.7	0.6
	Other Products	2.3	2.3	2.2	2.4	2.5	2.4	2.4	2.4	2.4	2.6	2.5	2.5	2.3	2.4	2.4	2.6	2.4
	Total Oil	15.8	15.8	15.7	16.4	16.4	16.4	17.1	16.9	16.4	17.4	17.4	17.4	15.8	16.4	16.8	17.4	16.6
World																		
	LPG	10.7	10.6	10.1	9.8	9.8	9.7	10.3	10.2	10.0	10.4	10.8	10.9	10.4	9.8	10.1	10.7	10.3
	Naphtha	6.9	7.0	6.9	6.5	6.4	6.3	6.8	6.7	6.5	6.5	7.0	7.2	6.9	6.4	6.7	6.9	6.7
	Gasoline	25.7	26.3	26.1	26.9	26.8	27.3	27.5	28.0	26.6	26.9	26.9	27.2	26.0	27.0	27.4	27.0	26.9
	Jet and Kerosene	7.3	7.4	7.5	7.8	7.7	7.8	8.0	8.1	7.8	8.0	7.7	8.0	7.4	7.8	8.0	7.9	7.8
	Diesel/Gasoil	28.1	29.6	29.3	29.2	29.5	29.6	29.5	29.8	29.4	30.0	29.9	29.3	29.0	29.4	29.6	29.8	29.4
	Fuel Oil	6.7	6.8	6.5	6.5	6.4	6.9	6.7	6.7	6.8	6.3	6.2	6.2	6.7	6.6	6.7	6.2	6.5
	Other Products	15.3	14.9	15.0	15.4	15.5	15.3	15.6	15.6	15.3	15.2	15.3	15.4	15.1	15.4	15.5	15.3	15.3
	Total Oil	100.7	102.6	101.3	102.1	102.1	102.9	104.4	105.1	102.4	103.4	103.9	104.2	101.5	102.4	104.0	103.8	102.9

Source: Wood Mackenzie, IEA, FlightAware, EIA, MarTrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

Table 33: Regional oil demand by product, 2025F

mbd		2025F												2025F				2025F
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	1Q	2Q	3Q	4Q	
United States																		
	LPG	1.3	1.2	1.1	0.9	0.9	0.9	1.0	0.9	1.0	1.1	1.2	1.2	1.2	0.9	1.0	1.2	1.1
	Naphtha	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.1
	Gasoline	8.2	8.4	8.5	8.7	8.7	9.0	8.8	9.1	8.5	8.7	8.6	8.3	8.4	8.8	8.8	8.5	8.6
	Jet and Kerosene	1.6	1.6	1.7	1.8	1.8	1.9	2.0	2.0	1.9	1.9	1.8	1.8	1.6	1.8	2.0	1.8	1.8
	Diesel/Gasoil	4.3	4.2	4.0	3.9	3.9	3.9	3.8	3.9	3.8	4.1	4.1	3.9	4.2	3.9	3.8	4.0	4.0
	Fuel Oil	0.4	0.3	0.3	0.2	0.2	0.4	0.4	0.4	0.4	0.4	0.2	0.3	0.3	0.3	0.4	0.3	0.3
	Other Products	4.4	3.9	4.0	4.3	4.2	4.2	4.4	4.5	4.3	4.3	4.4	4.4	4.1	4.3	4.4	4.4	4.3
	Total Oil	20.2	19.9	19.8	19.9	19.9	20.4	20.5	21.0	20.1	20.6	20.5	20.1	20.0	20.1	20.5	20.4	20.2
Europe																		
	LPG	1.0	1.1	1.0	1.1	1.0	1.0	1.1	1.0	1.0	1.0	1.0	1.1	1.0	1.0	1.1	1.0	1.0
	Naphtha	0.9	0.8	0.8	0.7	0.6	0.6	0.7	0.7	0.6	0.7	0.8	0.8	0.8	0.7	0.7	0.7	0.7
	Gasoline	2.0	2.1	2.0	2.2	2.2	2.3	2.3	2.3	2.2	2.2	2.1	2.2	2.1	2.2	2.3	2.2	2.2
	Jet and Kerosene	1.5	1.5	1.6	1.7	1.7	1.8	1.9	1.9	1.8	1.8	1.7	1.7	1.5	1.7	1.8	1.7	1.7
	Diesel/Gasoil	6.3	6.4	6.1	6.3	6.2	6.2	6.7	6.4	6.6	6.6	6.4	6.2	6.3	6.3	6.5	6.4	6.4
	Fuel Oil	0.8	0.9	0.8	0.8	0.9	0.9	0.8	0.8	0.8	0.7	0.7	0.7	0.8	0.8	0.8	0.7	0.8
	Other Products	1.3	1.3	1.4	1.5	1.4	1.4	1.5	1.4	1.4	1.4	1.3	1.2	1.3	1.4	1.5	1.3	1.4
	Total Oil	13.8	14.2	13.8	14.3	14.0	14.3	15.0	14.6	14.4	14.4	14.1	13.8	13.9	14.2	14.7	14.1	14.2
China																		
	LPG	2.5	2.5	2.3	2.4	2.4	2.4	2.6	2.6	2.5	2.5	2.8	2.7	2.4	2.4	2.6	2.7	2.5
	Naphtha	2.3	2.2	2.1	2.2	2.3	2.3	2.3	2.2	2.1	2.3	2.4	2.5	2.2	2.3	2.2	2.4	2.3
	Gasoline	3.8	3.8	3.6	4.1	4.1	4.0	4.1	4.1	4.0	4.1	4.0	4.0	3.7	4.0	4.0	4.0	4.0
	Jet and Kerosene	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.0	1.0	0.9	0.8	0.9	0.9	0.9	0.9	0.9
	Diesel/Gasoil	4.1	4.1	4.1	4.4	4.4	4.4	4.5	4.5	4.3	4.4	4.4	4.6	4.1	4.4	4.4	4.5	4.4
	Fuel Oil	0.7	0.7	0.7	0.7	0.7	0.7	0.6	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.6	0.7	0.7
	Other Products	2.2	2.3	2.1	2.3	2.3	2.4	2.4	2.3	2.2	2.5	2.4	2.5	2.2	2.3	2.3	2.5	2.3
	Total Oil	16.4	16.5	15.8	17.0	17.0	17.0	17.4	17.2	16.6	17.6	17.6	17.6	16.2	17.0	17.0	17.6	17.0
World																		
	LPG	10.8	10.7	10.0	10.0	10.0	9.8	10.4	10.2	10.0	10.5	10.8	10.9	10.5	9.9	10.2	10.8	10.3
	Naphtha	7.1	7.3	6.8	6.7	6.6	6.5	6.9	6.7	6.6	6.7	7.0	7.2	7.1	6.6	6.8	7.0	6.8
	Gasoline	25.6	26.1	25.9	26.9	26.8	27.2	27.3	27.8	26.6	26.8	26.8	27.0	25.9	27.0	27.2	26.9	26.7
	Jet and Kerosene	7.4	7.6	7.8	8.1	8.1	8.6	8.8	9.0	8.5	8.5	8.1	8.2	7.6	8.2	8.7	8.3	8.2
	Diesel/Gasoil	29.4	30.0	29.2	29.9	29.7	29.7	30.3	30.0	29.6	30.3	30.2	29.6	29.5	29.8	29.9	30.0	29.8
	Fuel Oil	6.7	6.9	6.6	6.5	6.5	6.9	6.7	6.8	6.8	6.3	6.1	6.2	6.7	6.6	6.7	6.2	6.6
	Other Products	15.3	15.0	15.0	15.4	15.5	15.4	15.7	15.6	15.3	15.2	15.4	15.5	15.1	15.4	15.5	15.4	15.4
	Total Oil	102.2	103.6	101.3	103.4	103.1	104.0	106.0	106.1	103.2	104.3	104.6	104.7	102.3	103.5	105.1	104.5	103.9

Source: Wood Mackenzie, IEA, FlightAware, EIA, MarITrace, Google, TomTom, US Department of Transportation, Various company reports and government statistics, J.P. Morgan

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